

Your company's cloud deployment is used by several teams and departments. You need to find an efficient way to ensure that the operating costs of the cloud deployment stays within the set budgetary limit specified by management. Which of these actions should you take to accomplish this?

- A) Use cloud resource labels and hierarchies

✗
- B) Set resource quotas and perform throttling

✓
- C) Manage inventory for on-premises systems

✗
- D) Use organizational policies for resource access

✗

Explanation

You should set resource quotas and perform throttling. Setting resource quotas and performing throttling needs to be done to ensure that cost overruns do not occur due to resource utilization that exceeds the specified budget. Using Google Cloud tools, you can perform automated throttling and capping of resource consumption as well as broadcast budget notification alerts on company communication channels, such as Slack. The most important capabilities for cost management on the cloud include:

- Maintaining costs within budgetary limits
- Finding ways to save on IT costs
- Performing forecasting for IT expenditure
- Maintaining consistency in IT costs and keep deviations limited

You would not use organizational policies for resource access in this scenario. You would use these policies to gain more control over your cloud costs. The purpose of using organizational policies is to specify permissions for users to perform spending on Google Cloud based on their role in the organizational hierarchy. This way, you control the ability of specific users to view costs and perform administrative functions using Cloud Identity and Access Management (Cloud IAM) for cloud resources.

You would not use cloud resource labels and hierarchies in this scenario. You can achieve greater accountability in your cloud deployment for financial governance by using cloud resource labels and hierarchies. Using resource hierarchies simplifies the process of allocating costs to specific departments to ensure accountability. This also provides insights into the return on investment (ROI) of the investment you have made on the cloud.

Managing inventory for on-premises systems is not a way of implementing budgetary control on a cloud deployment. An on-premises system and its inventory are independent of cloud resources. The financial management of on-premises systems is not related to the financial governance systems that need to be implemented to manage costs on a cloud deployment.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

You are a consultant for a stock trading company that is setting up its Google Cloud environment. You need to guide the team on some key issues that the company needs to be aware of when managing cloud environments. Which of these concerns should you specify? (Choose two.)

- A) Lack of on-premises servers

✗
- B) More overhead using cloud services

✗
- C) Lack of cost predictability

✓
- D) Less cloud usage transparency

✓
- E) Security of servers housing virtual machines

✗

Explanation

You should specify lack of cost predictability and less cloud usage transparency as key issues the company should be aware of. Lack of cost predictability can be an issue for companies managing cloud environments. A customer application may experience surges in user traffic alternated with periods of low levels of traffic. Due to the unpredictable nature of application usage, it can become difficult for a cloud administrator to specify resource limits. Setting resource limits too low can lead to application performance issues during peak usage, and provisioning more resources than required can lead to wasted capital. Less visibility and transparency in cloud usage leads to the wastage of cloud expenditure due to unutilized resources. By using cloud monitoring tools, a cloud consumer can identify resources that are unused and deprovision them so that they do not contribute to the overall billing. This helps reduce unnecessary costs for a cloud deployment.

The lack of on-premises servers is not a concern when managing a cloud environment because the set-up of an on-premises data center and its operation are not related to a cloud environment. To increase the computing power of a cloud deployment, you can perform scaling either manually or automatically, which provisions more virtual servers when required.

More overhead using cloud services is not a concern when managing a cloud environment. By using fully managed services on Google Cloud, you can reduce your operating costs by removing all the overhead that would be required for various tasks, which include resource provisioning, backups, updates, and patching. This reduction in costs allows you to better utilize your capital to grow your business.

The security of servers housing virtual machines (VMs) is not a concern when managing a cloud environment. This is the responsibility of the cloud service provider (CSP). As per the shared responsibility model of cloud environments, the customer is responsible for securing the data and applications that are hosted on the cloud, and the CSP secures the entire cloud infrastructure, which includes networking systems and physical servers that house VMs.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com >A guide to financial governance in the cloud](#)

Related Material

Your company has an on-premises data center that it uses to provide data processing services to customers. Due to the nature of the legacy system, there have been some business disruptions to the current users of specific services due to network traffic issues, anomalies, and malicious bot attacks.

What would be the best way to solve this issue?

- A) Use API management with Apigee

✓
- B) Build a document database with Firestore

✗

C) Move apps to the cloud using VMware engine

D) Create Kubernetes clusters with GKE

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✖

Explanation

You should use application programming interface (API) management with Apigee to solve this issue. API management with Apigee enables you to perform monitoring and analysis of API traffic to help mitigate key problems related to performance. It also allows you to build a layered defense system that includes distributed denial of service (DDoS) attack protection, bot handling, API guarding, as well as web application firewalls (WAFs). Apigee lets you monitor APIs and ensure your APIs are delivering expected performance and maintaining high availability for customers and developers. You can also identify any API issues and diagnose them easily using the monitoring systems for Apigee. API management with Apigee allows you to perform routing based on context, transform payloads, and perform API facades that are user specific. You can start by routing a small portion of the users to the modernized version of the service that uses APIs, and then based on the user feedback and experience, you can route all the users to the API-based modern system.

You would not create Kubernetes clusters with Google Kubernetes Engine (GKE) for this scenario. GKE is not used for API management, which is what this scenario needs. GKE allows you to create and manage Kubernetes clusters. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications. GKE provides detailed control over containers and nodes enabling you to specify CPU, RAM, storage, GPUs, and network configurations along with an Autopilot option for situations where there is not enough operational and developer support available.

You would not build a document database with Firestore. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

You would not move apps to the cloud using VMware engine. VMware Engine can be used to move applications that are based on VMware onto Google Cloud without any changes to existing applications. It provides all the necessary infrastructure and VMware licenses that allow you to have a software-defined data center (SDDC) on Google Cloud.

Objective:

Infrastructure & Application Modernization

Sub-Objective:

Understand the value of APIs

References:

Cloud.google.com > Apigee API Management

Related Material

Question #4 of 60

Question ID: 1495241

A company currently runs its operations using an on-premises data center. It is now consulting with you to modernize all its current applications using Google Cloud. What would be a key benefit of this transition?

A) Ensuring the cloud service provider performs application decoupling

B) Greater freedom for application modification and innovation

C) Automatic procurement of on-premises servers by the cloud service provider

D) The ability to make quick and permanent updates for fast app development

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Explanation

Modernizing applications using Google Cloud can allow software developers greater freedom for application modification and innovation. Decoupling is a key part of modernizing applications that allows for greater flexibility in making changes to parts of the application independently. One way to establish decoupling is using services like Pub/Sub. Pub/Sub is a middleware for messaging that enables applications to send messages to each other asynchronously. An application designated as a “publisher” application sends messages to a topic, and all the applications that have subscribed to that topic will receive the messages. This way, senders are decoupled from receivers in an architecture that is referred to as a “loosely-coupled” architecture, which forms the basis of a microservices-based architecture. Pub/Sub is useful for application developers that need to create a communication system between applications that are built independently of each other. Additionally, using application programming interfaces (APIs) to separate applications into multiple business services enables various domains of the organization, including the service owners and consumers to work at their own pace.

Ensuring the cloud service provider (CSP) performs application decoupling is not a key benefit. The CSP is not responsible for modifications or enhancements to a company's own applications. Those are the company's responsibility. When a company's infrastructure moves from an on-premises data center to a cloud deployment, the company's development team is still fully responsible for the source code. Cloud deployments follow a shared responsibility model where the CSP is responsible for the cloud infrastructure and its security, and the company is responsible for the data that is stored and used in the cloud.

Automatic procurement of on-premises servers by the CSP is not a key benefit. The CSP is not involved with procuring hardware for a company's on-premises data center. The company that owns the data center needs to maintain it and procure the necessary technology for the data center. The CSP manages the underlying infrastructure for its cloud system.

The ability to make quick and permanent updates for fast app development is not a key benefit. When developing modern applications, you use the DevOps approach to development, which states that changes should be gradual, reversible, and fully documented. This approach provides greater reliability and stability for systems and can improve the mean time to recovery (MTTR) during a service disruption. A key DevOps practice is continuous integration and continuous delivery (CI/CD), which specifies that a controlled and systematic approach needs to be used when making changes to ensure system stability and avoiding configuration drift.

Objective:

Infrastructure & Application Modernization

Sub-Objective:

Understand modernizing applications with Google Cloud

References:

Inthecloud.withgoogle.com > CIO Guide to Application Modernization > Understanding and Decoupling Applications

Related Material

Question #5 of 60

Question ID: 1495939

Your organization needs to find a cybersecurity solution that can handle its massive size and complexity in a unified system. Which Google Cloud solution should you use to monitor massive amounts of data and identify cybersecurity threats?

A) Cloud Trace

B) Chronicle

C) Looker

D) Cloud Endpoints

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Explanation

You should use Chronicle, which is Google's cloud-native Security Information and Event Management (SIEM) system. Chronicle is created to find cyberthreats by analyzing large amounts of data. It works by ingesting an organization's data into a container and then integrating it with Google's own data. The data is then combined and linked into a single timeline. Finally, the data is worked on using application programming interfaces (APIs) to investigate, detect, and hunt cyberthreats. Some current cybersecurity challenges and threats you should know about include the following:

- Phishing – This is a social engineering attack where a user can be manipulated to reveal sensitive personal data and credentials through a fraudulent email. Google provides protection against phishing by using Universal 2nd Factor (U2F) compatible security keys in place of One-Time Transaction Password (OTP)-based second factor authentication. Google's Titan Security Keys provides protection from phishing using two-factor authentication (2FA) devices for high-value user protection.
- Malware – This comprises any software or program that is designed to cause damage to a computer system and steal data. Malware includes viruses, trojans, ransomware, and spyware. Google Cloud provides VirusTotal and Google Cloud Threat Intelligence for Chronicle for monitoring and responding to malware.
- Unsecured third-party systems – Third-party software can increase cybersecurity risks. Third-party software can have security issues in it, which can compromise an organization's data security and make the organization vulnerable to cyberattacks. There needs to be a comprehensive security architecture in place within a company's IT processes to check third-party software and monitor it on a regular basis to ensure there are no security issues.
- Physical damage – This comprises any damage that can be done to physical devices, such as servers, networking components, and disk drives. To protect its infrastructure from physical damage, Google uses several layers of physical security at its data centers, including biometrics, metal detectors, vehicle barriers, cameras, and intrusion detection systems that are laser-based.

Other cybersecurity threats include unsafe endpoints, formjacking, cryptojacking, Internet of Things (IoT) attacks like botnets, and patching issues.

You would not use Cloud Trace in this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights into its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

You would not use Cloud Endpoints in this scenario. Cloud Endpoints provide a system for managing application programming interfaces (APIs) that enables monitoring, analyzing, and securing APIs as well as specifying quotas for them. You can monitor the key operational metrics for the APIs using Google Cloud Console and gain usage insights with BigQuery, Cloud Trace, and Cloud Logging. You can also utilize the Cloud Endpoints Portal to create a website that API users can access to find documentation for the APIs and interact with them.

You would not use Looker in this scenario. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It can also be embedded into other applications to provide embedded analytics. Looker can be used to create alerts and proactively send timely insights related to various business data. It also has dashboards that are interactive, which allow you to analyze data in real time through a trusted data model.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Chronicle: Google's Cloud-Native SIEM > the Security Operations suite for the modern SOC](#)
[Cloud.google.com > Google Infrastructure Security Design Overview](#)

Related Material

Question #6 of 60

Question ID: 1495186

An e-commerce company is expanding its workforce and is working on performing data analysis to find ways to grow its business. You need to use a suitable service to improve decision making and productivity in the organization by providing key insights to a growing number of users. Which tool should you use for this requirement?

- A) Cloud Trace

✖
- B) Dataproc

✖
- C) Dataflow

✖
- D) Looker

✔

Explanation

You should use Looker. Looker is a platform for business intelligence (BI) that allows you to access intelligent data for making better business decisions. It can be embedded into other applications for providing embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow for analyzing data in real-time through a trusted data model.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Dataflow for this scenario. Dataflow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

You would not use Cloud Trace in this scenario as it is not for BI, but a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Looker](#)

Related Material

Question #7 of 60

Question ID: 1495224

You need to help your company choose an appropriate cloud architecture for its operations. Which of these is an advantage of adopting a hybrid cloud architecture?

- A) Safeguarding all applications and data from public clouds

✖
- B) Ending data center costs by migrating infrastructure to a public cloud

✖
- C) Maximizing security of sensitive data through on-premises storage

✔
- D) Availing services from multiple public cloud service providers

✖

Explanation

Maximizing security of sensitive data through on-premises storage is an advantage of adopting a hybrid cloud architecture. By using a hybrid cloud architecture, you can provision resources from a cloud service provider (CSP) and not have to procure the same computing resources for the data center. In a hybrid cloud system, a company uses its own data center along with a public cloud system to achieve its business goals. A hybrid cloud also always contains a private cloud. Some key advantages of adopting a hybrid cloud architecture include greater agility in handling usage bursts, leveraging new and advanced analytics technology, and speeding up time-to-market and reducing cycle times for new applications.

Availing services from multiple public CSPs is not a feature of hybrid cloud architectures but of multi-cloud architectures. A multi-cloud deployment involves using multiple public CSPs where each public cloud service has a different function for the company. This allows a company to use one public cloud for database services and another one for analytics. A company can then use a preferred cloud provider for specific services. Remember that using multiple public CSPs indicates a multi-cloud infrastructure.

Safeguarding all applications and data from public clouds is not a feature of hybrid cloud architectures but of private cloud architectures. In this, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premises data center. This system does not leverage systems that are a part of a public cloud service offering.

Ending data center costs by migrating infrastructure to a public cloud is not an aspect of hybrid cloud architectures but of public cloud architectures. This infrastructure includes applications that are fully functional in a cloud and have no on-premises portions. This infrastructure option may not be suitable for situations where sensitive data needs to be stored locally and not on a public medium.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #8 of 60

Question ID: 1495932

A company wants to find a way to find anomalies in shopping transactions of users that browse its site and mobile app through a secure API-based system that is fully managed and scalable. Which Google Cloud service should you use for this?

- A) Looker**
- B) Payment Gateway**
- C) Dataproc**
- D) Firestore**
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Explanation

You should use Payment Gateway for this scenario. With Payment Gateway, you can perform processing of payments that are application programming interface (API)-based. It can scale up to cater to unpredictable surges in usage as well as be extended to third-party insights and anomaly finding tools through APIs. It also allows for payment processing across multiple regions of Google Clouds global infrastructure. Payment Gateway also has built-in tools that allow you to monitor, debug, and trace API-based purchases with high availability and less overhead. You can also export data of transactions into a project on Google Cloud using an export API.

You would not use Firestore for this scenario. Firestore is a serverless document database which is scalable and fully managed. It enables fast development for applications for the internet, mobile, and Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Looker for this scenario. Looker is a platform for business intelligence (BI) that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications to provide embedded analytics. Looker can be used to create alerts to proactively send timely insights related to various business data. It also has dashboards that are interactive, which allow you to analyze data in real-time through a trusted data model.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > Payment Gateway](#)

Related Material

Question #9 of 60

Question ID: 1495940

Your company has multiple teams that have hundreds of employees in them. You need a centralized way to manage user access to cloud resources. Which of these systems would you need to use to authorize which users can access specific Google Cloud resources?

- A) Cloud IAM**
- B) Encryption at rest**
- C) Secret Manager**
- D) Billing APIs**
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Explanation

You would use Cloud Identity and Access Management (IAM) for this scenario. Cloud IAM enables you to manage cloud resources centrally with granular access control and visibility. With IAM, you can grant access at the resource level, such as providing the role of Subscriber to a user for a Pub/Sub topic. This way, IAM allows you to grant roles in addition to Owner, Editor, and Viewer roles. IAM also provides Recommender that can enable you to detect overly permissive access policies and rightsize them. With IAM, resource access is controlled using contextual attributes, such as IP address or resource type. You can perform IAM policy management and creation using the Google Cloud Console, programmatically using IAM methods, or by command-line access using the gcloud tool. Google Accounts are also supported by IAM. You can utilize Cloud Identity with IAM to synchronize user accounts across multiple projects. IAM also provides workforce identity federation that enables you to perform authentication and authorization of a workforce using external identity providers (IdPs). IAM can provide a full audit history that includes granting, removal, and delegation of permissions for compliance needs.

You would not use Secret Manager for this scenario. You use Secret Manager for storing sensitive data like passwords, application programming interface (API) keys, and certificates securely. With Secret Manager, you have a single source of truth for accessing, managing, and auditing secrets on a Google Cloud deployment. Secret Manager provides lifecycle management with versioning for pinning requests to the newest versions of secrets, as well as the ability to use Cloud Functions to automate the process of rotating secrets. It also allows you to select regions for the replication of your secrets data. Secret Manager is integrated with Cloud IAM for controlling access to secrets.

You would not use encryption at rest in this scenario. Encryption at rest of stored data is a way of implementing security for data stored on the cloud infrastructure. Google utilizes multiple layers of encryption for the protection of data at rest. Google's storage infrastructure encrypts all user data by default before it is stored on physical storage. This way, encryption is performed by Google Cloud at the storage or application layer of the cloud infrastructure. Google has several storage services like Colossus and Cloud Spanner, as well as a key management service.

You would not use billing APIs for this scenario. You can use billing APIs to access various tools created by other vendors to manage project bills programmatically and obtain data on all stock keeping units (SKUs) in their repository and related metadata. This is a way of ensuring clarity for financial governance on a cloud deployment. Other methods include using intelligence recommendations for performing rightsizing of computing resources. This helps ensure that costs are minimized by ensuring that virtual machine (VM) usage is optimized.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Cloud Identity and Access Management \(IAM\)](#)

Related Material

Question #10 of 60

Question ID: 1495896

Which of these definitions best describes the meaning of the key term “Launch and iterate” in the context of revolutionizing business using Google’s cloud technology?

- A)** A focus on creating solutions that make improvements by order of 10 times
- B)** A focus on creating a solution that makes improvements by 10 percent
- C) A focus on finding the right solution through continuous experimentation**
- D)** A focus on building a perfect solution right at the very beginning

Explanation

The meaning of the key term “Launch and iterate” is best described as “A focus on finding the right solution through continuous experimentation”. This approach encourages organizations to constantly try out new systems and technologies to arrive at a solution that works rather than attempting to create a perfect solution from the start.

“A focus on building a perfect solution right at the very beginning” is not the meaning of launch and iterate. Finding the perfect solution at the start is not recommended because continuous experimentation with new technologies can help create the right solutions.

“A focus on creating solutions that make improvements by order of 10 times” and “A focus on creating a solution that makes improvements by 10 percent” are both incorrect. “Think 10x” is a key term that focuses on creating solutions that make improvements 10 times over.

For the Google Cloud Digital Leader exam, you need to pay special attention to definitions of certain key terms used across the exam. Your correct answer choice will depend greatly on your familiarity with these terms. Going through the associated reference material is also recommended.

- Cloud – This comprises a data center network that provides storage and computation for information using the Internet. The cloud includes all the systems that make this possible, including computers, software applications, networking systems, and security technology.
 - Cloud technology or cloud computing – This refers to the systems and procedures required for storing, managing, and providing access to data sent across the cloud. This data is separate from the data that resides on a user’s personal computer’s hard drive.
 - Data – In the context of Google Cloud, data refers to information that is of value to a company. Data can include audio and video content, textual information in an email, numbers in a spreadsheet file, imagery, and employee ideas. Data can include information that is internal or external to a company.
 - Digital transformation – This occurs when a company utilizes new technology to change the design and definition of its relationships with its partners, customers, and employees. Digital transformation involves using new digital systems including private, public, and hybrid clouds to change existing business culture, work processes, and customer experiences with an aim to meet the changing dynamics of businesses and markets.
- Computing – This is the ability of a machine to perform storage, processing, retrieval, comparisons, and analysis of information. It also includes the machine's ability to automate tasks using computer applications.
 - Compute power – This refers to the speed of a computer in processing data.
 - Think 10x – This is also referred to as “generating big ideas” and involves looking for solutions to make improvements by an order of 10 times rather than by 10 percent.
 - Launch and iterate – This is also referred to as “continuous learning” and involves a perspective of finding the right solution through continuous experimentation rather than attempting to craft a perfect system right at the beginning.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Service.google.com > Cloud Digital Leader Study Guide](#)

Related Material

Question #11 of 60

Question ID: 1495239

Your company is a software development firm that has been using a legacy on-premises system and now wants to leverage cloud technology to reduce bugs caused by differences in the development and production environments. What kind of change pattern for application modernization should you opt for in this use case?

- A) Perform containerization of all services gradually**
- B)** Use a manual release process for all systems
- C)** Keep specific personnel dedicated to testing tasks
- D)** Combine multiple services into a single monolith

Explanation

You should perform containerization of all services gradually. When a development environment is too different from an existing production environment, there can be various bugs and errors generated. A solution to this is to containerize all services gradually over time. This creates a greater level of control and predictability over the environments for development, production, and staging. You start by containerizing a single service. After you have gained confidence using certain processes and tools for containerization, you can proceed to containerize all services. This is a key aspect of application modernization.

You would not use a manual release process for all systems. A release process for software that is performed manually is prone to errors and is slower than an automated system. A key DevOps practice is continuous integration and continuous delivery (CI/CD), which specifies that a controlled and systematic approach needs to be used when making changes to ensure system stability and avoiding configuration drift. A public cloud system offers features like CI/CD for rapid and automated development and updates for applications. You would first use a simple CI model and then employ CD. After some time, you would apply the CI/CD model to all existing services. Then you can add additional stages to your CI/CD pipeline gradually.

You would not keep specific personnel dedicated to testing tasks. You would instead opt for automated testing systems. Manual processes for testing are prone to human errors. For this reason, when performing application modernization, an automated approach is recommended. By utilizing a basic CI/CD pipeline for your containers, you can have an automated system for testing and deploying your software.

You would not combine multiple services into a single monolith. Decoupling is a key part of modernizing applications that allows for greater flexibility in making changes to parts of the application independently. This is a form of functional decomposition that breaks down an existing legacy or “monolith” application into independent applications. One way to establish decoupling is using services like Pub/Sub. Pub/Sub is a middleware for messaging that enables applications to send messages to each other asynchronously. An application designated as a “publisher” application sends messages to a topic, and all the applications that have subscribed to that topic will receive the messages. This way, senders are decoupled from receivers in an architecture that is referred to as a “loosely-coupled” architecture, which forms the basis of a microservices-based architecture.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Related Material

Question #12 of 60

Question ID: 1495898

Your company is considering using a hybrid cloud setup where you will use several public clouds and your existing on-premises servers.

Which cloud migration method would you use if you need to refactor an existing application and modernize it before moving it to a cloud deployment?

- A) Improve and move

B) Lift, tinker, and shift

C) Rip and replace

D) Lift and shift
- ✔

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Explanation

You would use the “Improve and move” method in this scenario. This method involves taking an existing application and first refactoring and modernizing it. Then the application is moved to a public cloud system. This movement to a public cloud is an ideal situation for companies that want to create new products and services using new technologies.

You would not use the “Lift and shift” method in this scenario. This method involves the migration of an application that was previously in a private cloud environment to a public cloud without making any key modifications. This approach is used by companies that want minimal disruption to their operations, and want to migrate to the cloud while keeping their operating costs as low as possible.

You would not use the “Rip and replace” method in this scenario. This method involves replacing an existing application completely with a new cloud-based version. This approach is used for situations where modernizing the existing application is simply not a feasible alternative due to cost or technology related limitations.

You would not use the “Lift, tinker, and shift” method in this scenario. This method is for situations where you need to modify parts of an application to gain the benefits of a cloud deployment without changing the underlying core of the application.

When considering migration strategies for a hybrid cloud deployment, you also need to be aware of the utilities of a hybrid cloud system:

- Well-paced modernization – A hybrid cloud approach allows you to migrate your applications to a cloud at a speed that is comfortable for your organization. This way you can change the IT infrastructure of your organization over a period of time.
- Regulatory compliance – Each industry has certain stipulations that specify where an application is allowed to operate. Using a hybrid cloud deployment ensures that you keep the use of your application and data within regulatory compliance laws.
- On-premises applications – Some of your applications may need to be run on mainframe computers on-premises. Moving these systems to a cloud deployment may not be feasible. For such regulated applications, a hybrid cloud deployment may be suitable.
- Applications at edge locations – Some industries and applications require the use of edge computing to provide low latency. This could be for systems like retail kiosks or telecommunication networks. Hybrid cloud systems allow you to run such specific applications at remotely placed edge locations.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is a Hybrid Cloud? > How-to hybrid](#)

Related Material

Question #13 of 60

Question ID: 1495245

Your company is looking for a serverless platform that is fully managed for running containers individually. What is the most efficient way to write code for an application and deploy it as a container with no additional overhead required?

- A) Cloud Functions

B) Cloud Run

C) App Engine

D) Compute Engine
- ✖

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✖

Explanation

You should use Cloud Run for this scenario. You can use Cloud Run for situations where you need to perform deployment of containerized applications that are written in your preferred coding language. Cloud Run provides support for Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS), and websockets. You can use Cloud Run for creating application programming interfaces (APIs), applications for processing data, webhooks, and websites. With Cloud Run, code can be packaged into a container and simply deployed without needing infrastructural scaling. The apps containerized with Cloud Run can be written in Ruby, Python, Go, Java, Node.js, or .NET.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources like Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). An example is running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not use Compute Engine for this scenario. Compute Engine provides virtual machine (VM) computing that has scalability and flexibility. It allows you to utilize Cloud Tensor Processing Units (TPUs), Graphics Processing Units (GPUs), and CPUs to perform analytics and processing on a large scale using Google’s technological infrastructure.

You would not use App Engine for this scenario. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine, you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning. App Engine provides support for web application development using programming languages like Python, Ruby, C#, Node.js, Go, and PHP.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Cloud Run](#)

Related Material

Question #14 of 60

Question ID: 1495176

A company currently organizes training for upskilling its employees and relies on the assistance of third-party companies to run its training sessions. How can the company move from this phase to a transformational cloud adoption maturity phase?

- A) Reduce training within the company and increase training from third parties

✖
- B) Reduce third-party involvement to a minimal level and increase knowledge sharing through peers

✔
- C) Encourage self-teaching for employees and bring in more third parties for training

✖
- D) Reduce peer-based knowledge sharing and bring in third-party support for learning

✖

Explanation

The correct approach to move the company to a transformational phase in the Cloud Maturity Scale is to reduce third-party involvement to a minimal level and increase knowledge sharing through peers. A key aspect of the transformational phase in the Cloud Maturity Scale is increasing knowledge sharing and lateral awareness while ensuring individuals and teams can make more autonomous decisions.

The other options are incorrect as a key aspect of the transformational phase is to reduce the involvement of third parties to only augment what is done internally in the company through raised awareness and peer-based knowledge sharing.

A company's Cloud Maturity Scale is formed when the four themes are evaluated using the three phases:

- Tactical
 - Learn – Learning is mainly self-taught, and there is reliance on third parties for knowledge.
 - Lead – Each function’s team has a leader, and there is a strong project manager.
 - Scale – Changes are risky, slow, and have high operational overhead.
 - Secure – The public Internet is avoided for security reasons, and the private network is preferred.
- Strategic
 - Learn – Learning is organized through training and can have assistance from third parties.
 - Lead – There are cloud teams that operate across functions.
 - Scale – Scaling is templatized, which maintains proper governance without the need for manual supervision.
 - Secure – The network is a hybrid one, and there is a centralized system for identity management.
- Transformational
 - Learn – Learning occurs through peers and knowledge sharing. Third parties only augment the ongoing learning culture.
 - Lead – There is more autonomy for teams and there are feature teams that operate across functions.
 - Scale – Changes are low risk, constant, and can be easily remediated, if required.
 - Secure – This is a low trust model where only the appropriate devices, users, and services are trusted.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > The Google Cloud Adoption Framework](#)

Related Material

Question #15 of 60

Question ID: 1497127

You need to recommend a Google product to a company that wants to enhance its applications by incorporating intelligent components that are plug-and-play. For this, the company needs to create, deploy, and scale its machine learning models quickly using a single platform. Which Google product should you recommend for this scenario?

- A) Network Intelligence Center

✖
- B) Cloud Life Sciences

✖
- C) Memorystore

✖
- D) Vertex AI

✔

Explanation

You should recommend Vertex Artificial Intelligence (AI) for this scenario. Vertex AI provides a singular application programming interface (API) platform for building and scaling machine learning (ML) models swiftly with features for custom tooling and pre-training. It allows you to manage the AI and ML development lifecycle for a company. Vertex AI has various services and tools within it, such as Automated Machine Learning (AutoML), which is a system that allows you to perform training for ML models without the need for coding. Vertex AI also allows you to create ML models that are advanced using its custom model tooling. It has Machine Learning Operations (MLOps) tools that enable ML model maintenance using the Vertex AI Pipelines, which streamlines the process of running pipelines for ML, and the Vertex Feature Store, which you can use to serve and use AI technologies as features for your ML applications. ML is the science of developing intelligent systems and applications that can “learn” from existing data provided to it and it can then provide services that are intuitive, dynamic, and adaptive. ML is a subset of AI that involves methods for making a machine mimic the ability of a human to make decisions and solve problems. This is done using deep learning and neural networks that enable the ML system to learn continuously. It removes the need to program the ML system explicitly by providing it with large quantities of data.

You would not recommend Network Intelligence Center for this scenario. Network Intelligence Center is used to monitor, verify, and optimize a network. It can be used for Google Cloud, on-premises, and multi-cloud systems. Network Intelligence Center reduces the risk of outages occurring and ensures greater levels of compliance with regulations and security.

You would not recommend Memorystore for this scenario. Memorystore is a fully managed service that provides in-memory storage of data with less than a millisecond of response time to access the data. It does this by deploying distributed caches for applications. Memorystore helps migrate the cache layer to the cloud without needing to modify existing application code.

You would not recommend Cloud Life Sciences for this scenario. Cloud Life Sciences enables you to manage, process, and transform data related to life sciences. This data can include genomics and biomedical data. Cloud Life Sciences provides workflows that are containerized and allow for the processing, analysis, and annotations of life sciences data with scalability.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Vertex AI](#)

Related Material

Question #16 of 60

Question ID: 1495910

For your company's data-driven operations, you are looking for a way to organize all the organizational data in the company by various business domains, which include customer support, human resources, and sales, and use a decentralized structure in a way that provides federated data ownership. Which of these options would be the most appropriate for this requirement?

- A) Data lake

B) Data lakehouse

C) Data warehouse

D) Data mesh
- ✖

✖

✖

✔

Explanation

A data mesh would be the most appropriate choice for this requirement because a data mesh provides a decentralized system architecture for organizing data by business domains. This ensures greater ownership for the people that produce each dataset. Producers can then specify policies for data governance. These policies enable greater self-service by users across the organization through the emphasis on clear documentation, quality concerns, and access systems. Using a data mesh requires cloud adoption for the necessary scalability and achievement of data management goals. Using a data mesh removes operational hurdles associated with more centralized and monolithic data systems. A data mesh provides federated ownership of data and allows data access by multiple teams through the use of application programming interfaces (APIs).

You would not use a data lakehouse in this scenario. A data lakehouse is a data platform that combines data lakes and data warehouse technologies in a way that uses the tools and management features of data warehouses and merges them with the flexibility of storing unstructured data in data lakes. You take away the overhead that is involved with using data warehouses and data lakes by using a data lakehouse. A data lakehouse provides a single point of access to both real-time and batch data, and also provides ways to implement data governance.

You would not use a data lake in this scenario. A data lake allows you to store structured, semi-structured, and unstructured data from multiple sources in a single place. This data can come from a variety of sources, including multimedia, social media, hard disks, servers, Internet of Things (IoT) devices, clickstreams, databases, and transactional information. A data lake provides security and scalability and enables companies to perform ingestion of data from various sources, including edge computing, cloud, or on-premises systems. The data can be stored in its original format and processed as a batch or in real-time. This data can then be analyzed using tools like Python, SQL, and R.

You would not use a data warehouse in this scenario. A data warehouse is a massive repository that contains a company's data gathered from various different sources. This data includes current data as well as data that is historical. A data warehouse is used for business intelligence (BI) activities like complex analytics, which can lead to making informed management decisions. The sources of data for a data warehouse include customer relationship management (CRM) systems, automation systems for marketing, applications for transactions, and log files from applications. A data warehouse can be used for performing analysis in an ad hoc manner or for creating custom reports.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Services.google.com > Build a modern, unified analytics data platform > Emerging Trends](#)

Related Material

Question #17 of 60

Question ID: 1495226

Your company has a two-tier application with a web-based frontend for interacting with users and an RDBMS backend that stores user data. You need to find a way to gradually move your systems from on-premises to a cloud deployment. How should you implement a tiered hybrid cloud deployment?

- A) Migrate backend applications to a public cloud

B) Store data that is restricted by regulations on a public cloud

C) Use a private cloud for performing changes to frontends

D) Deploy frontend applications to a public cloud
- ✖

✖

✖

✔

Explanation

You should deploy frontend applications to a public cloud. Frontend applications require frequent changes and updates as they face users and are performance centric. By moving frontend applications to a public cloud, you can have an automated continuous integration and continuous delivery (CI/CD) system for efficient deployment of updates to applications.

You would not migrate backend applications to a public cloud. Backend applications can have various system dependencies that are very complex, and the task of migrating these applications to a public cloud can be an involved and challenging process. By focusing on moving the frontend applications to the cloud first, you can perform cloud integration faster and utilize cloud-native services for increased availability and resiliency for the system.

You would not store data that is restricted by regulations on a public cloud. Any data that is subject to regulatory restrictions should ideally be stored in an on-premises environment. This approach ensures greater security of the data.

You would not use a private cloud for performing changes to frontends. This is because a public cloud system offers features like CI/CD for rapid and automated development and updates for applications. Additionally, cloud services, including Cloud Internet of Things (IoT) and Cloud Content Delivery Network (CDN), can be useful for creating user interfaces, data ingestion from IoT sources, and creating application programming interfaces (APIs).

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #18 of 60

Question ID: 1497138

You work for an online shopping company that allows customers to write product reviews as text on the shopping website and app. You want to enable your developers to extract a user's sentiment from blocks of text from the user's review to better understand customer experiences. Which of these technologies would you use for that purpose?

- A) Cloud Translation

✖
- B) Cloud Trace

✖
- C) Cloud Vision

✖
- D) Cloud Natural Language API

✔

Explanation

You would use Cloud Natural Language application programming interface (API) for this scenario. Cloud Natural Language API or Natural Language Artificial Intelligence (AI) enables you to gain understanding of natural language using API. This can help you identify the following aspects in a block of text:

- Key entities
- Positive or negative sentiments
- Language used
- Syntax including dependency trees and speech parts

The Cloud Natural Language API can be used by developers through API calls or through reference to a document that may be stored on Cloud Storage.

Cloud Translation is not the appropriate choice for this scenario. Cloud Translation or Translation AI performs the translation of text between languages using Representational State Transfer (RESTful) APIs. You can use Cloud Translation to translate text from English to Spanish to any other language and vice versa. By using Cloud Translation API, you can provide automatic text translation features for an app or Website.

Cloud Vision is not the appropriate choice for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an API. It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images. Cloud Vision allows you to keep moderation for image content, use analysis of image sentiment for marketing strategies, and create metadata for an image catalog. It also allows you to perform image analysis for images uploaded on requests and can be integrated with Cloud Storage.

Cloud Trace is not the appropriate choice for this scenario. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Natural Language AI](#)

Related Material

Question #19 of 60

Question ID: 1495292

Your company has various systems and services running on Google Cloud. Each of these generates events. You need a way to create alerts whenever a specific error event on the system is generated. Which Google Cloud service should you use to accomplish this?

- A) Cloud Vision

✖
- B) Cloud Armor

✖
- C) Cloud Logging

✔
- D) Cloud Spanner

✖

Explanation

You should use Cloud Logging for this scenario. Cloud Logging is a log management service that is fully managed. It allows you to store, search, and analyze exabytes of log data, as well as set up alerts and notifications. Cloud Logging has Logs Explorer, which lets you use flexible queries for searching, sorting, and analyzing logs. You can also utilize log buckets to store and process log data based on regional compliance requirements. Cloud Logging also has an Error Reporting feature that performs a smart collection of errors in logs into error groups. Cloud Logging also allows your security teams to keep audit trails using its Cloud Audit Logs feature, which provides greater transparency in activities related to administration. Cloud Logging uses the Cloud Logging Application Programming Interface (API) to collect log entries passed through a Logs Router. The Logs Router uses existing filters to identify the log entries meant for exclusion, ingestion, or importing.

You would not use Cloud Spanner for this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner for the provision of a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an API. It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images.

You would not use Cloud Armor for this scenario. Cloud Armor is a service that protects websites and software applications from threats like web attacks and distributed denial of service (DDoS) attacks. It offers adaptive protection by detecting and mitigating large numbers of Layer 7 DDoS incidents using a locally trained machine learning (ML) system.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Logging](#)

Related Material

Question #20 of 60

Question ID: 1497155

Your company has recently moved to a cloud deployment and is struggling to manage its growing cloud costs. You are asked to suggest some ways to get costs under control. Which of these should you suggest? (Choose two.)

- A) Run all VMs continuously

✖
- B) Set usage quotas for resources

✔
- C) Store all data with the standard class

✖
- D) Create alerts for budget overruns

✔
- E) Duplicate data across buckets

✖

Explanation

You should suggest the company set usage quotas for resources and create alerts for budget overruns. Some controls available to companies for keeping cloud costs and usage under control include budgetary alerts, usage quotas, and company permissions. Setting resource quotas and performing throttling must ensure that cost overruns do not occur due to resource utilization exceeding the specified budgets. Using Google Cloud tools, you can perform automated throttling, capping of resource consumption, and broadcast budget notification alerts on company communication channels, such as Slack.

You would not store all data with the standard class. This action can cause wasted capital due to overspending on storage. Google Cloud Storage offers various storage classes, which include standard, nearline, coldline, and archival. By moving data to less expensive storage classes, such as the archival class, you can save on costs for storing data that is accessed rarely. By using a lifecycle policy, you can automatically ensure that data objects change their storage class based on conditions you specify, which can be the age of the object.

You would not run all virtual machines (VMs) continuously. Using auto start and stop for VMs can help reduce Compute Engine's operating costs. By deactivating VMs during non-business hours rather than allowing VMs to run continuously, you can gain significant cost savings. You can use a serverless system to automate the process of VM shutdowns by using Cloud Pub/Sub, Cloud Scheduler, and Cloud Functions.

You would not duplicate data across buckets. Duplicating data causes a waste of storage and capital. Some ways you can prevent creating duplicate data is by using multi-region buckets and setting object versioning on Cloud Storage correctly. By specifying an object versioning policy, you can maintain a suitable number of copies of data, and you can also use the bucket lock version to protect from accidental deletion of objects.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #21 of 60

Question ID: 1495222

The company you work for wants to find more secure and cost-effective methods of managing its systems. You are looking to select the appropriate cloud deployment for this. Which of these deployment configurations uses a hybrid cloud infrastructure?

- A) Using one public cloud service provider for a web frontend and a different one for the backend

✖
- B) Storing all organizational data and applications in an on-premises data center

✖
- C) Using an on-premises data center for storing raw data along with a public cloud for data analytics

✔
- D) Using SaaS from one cloud provider and PaaS from another

✖

Explanation

Using an on-premises data center for storing raw data along with a public cloud for data analytics is an example of a hybrid cloud infrastructure. In a hybrid cloud system, a company uses its own data center along with a public cloud system to achieve its business goals. A hybrid cloud also always contains a private cloud. A hybrid cloud deployment allows a company to run its applications on-premises and use the public cloud for bursting when user demands go beyond the limits of the data center.

Using one public cloud service provider (CSP) for a web frontend and a different one for the backend is not an example of a hybrid cloud deployment but of a multi-cloud deployment. A multi-cloud deployment involves using multiple public CSPs where each public cloud services a different function for the company. A company can then use one public cloud for database services and another one for analytics. This lets a company use its preferred cloud provider for specific services. Remember that using multiple public cloud service providers indicates a multi-cloud infrastructure.

Using Software as a Service (SaaS) from one cloud provider and Platform as a Service (PaaS) from another is not an example of a hybrid cloud infrastructure but of a multi-cloud infrastructure. This is because both SaaS and PaaS are public cloud offerings, and using several public clouds is a feature of multi-cloud deployments. The PaaS model allows consumers to install their own applications onto a cloud platform. The provider supplies the necessary operating system and physical hardware. This allows consumers to quickly deploy their own applications without the burden of first acquiring the necessary physical resources and operating system(s) to make it happen. The SaaS model covers the use of software applications by consumers who are running on a provider's cloud infrastructure. These applications can be accessed using thin client interfaces, including mobile devices and web browsers. An example of this is email accessible from a web browser, such as Gmail. The advantage of this model is that a consumer can utilize the application without the need to physically install the application or be concerned about the infrastructure needed to run it.

Storing all organizational data and applications in an on-premises data center is not an example of a hybrid cloud infrastructure but of a private cloud infrastructure. In this infrastructure, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premise data center.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #22 of 60

Question ID: 1495297

You need to store the events generated by various Google Cloud services in buckets that need to be localized to specific regions for regulatory and compliance requirements. Which Google Cloud service would be the most appropriate choice for this?

- A) Cloud Logging

✓
- B) Cloud Endpoints

✗
- C) Cloud Trace

✗
- D) Cloud Profiler

✗

Explanation

You would use Cloud Logging for this scenario. Cloud Logging is a log management service that is fully managed. It allows you to store, search, and analyze exabytes of log data as well as set up alerts and notifications. Cloud Logging has Logs Explorer, which lets you use flexible queries for searching, sorting, and analyzing logs. You can also utilize log buckets to store and process log data based on regional compliance requirements. Cloud Logging also has an Error Reporting feature that performs a smart collection of errors in logs into error groups. Cloud Logging also allows your security teams to keep audit trails using its Cloud Audit Logs feature, which provides greater transparency in activities related to administration. Cloud Logging uses the Cloud Logging Application Programming Interface (API) for collecting log entries, which are passed through a Logs Router. The Logs Router uses existing filters to identify the log entries meant for exclusion, ingestion, or import.

You would not use Cloud Profiler for this scenario. Cloud Profiler is used to find and remove any possible performance issues from your applications currently in production. This is accomplished by Cloud Profiler regularly profiling the consumption of resources across your production applications. Cloud Profiler creates an analysis of the performance of an application by collecting data related to CPU usage and memory allocation.

You would not use Cloud Trace for this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights into its performance.

You would not use Cloud Endpoints for this scenario. Cloud Endpoints provide a system for managing APIs that enables monitoring, analyzing, and securing APIs as well as specifying quotas for them. You can monitor the key operational metrics for the APIs using Google Cloud Console and gain usage insights with BigQuery, Cloud Trace, and Cloud Logging.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Logging](#)

Related Material

Question #23 of 60

Question ID: 1497140

You need to create a system that can provide intuitive and high quality conversations for customers through voicebots and chatbots. What Google technology would you select for this need?

- A) Visual Inspection API

✗
- B) Document AI

✗
- C) Dialogflow

✓
- D) Media Translation API

✗

Explanation

You would use Dialogflow in this scenario. Dialogflow provides a suite for the development of conversational apps with voice and text capabilities, which include voicebots as well as chatbots. Dialogflow provides cross-platform connectivity to existing apps for the Internet of Things (IoT), iOS, or Android, and for telephony platforms, including Cisco, Genesys, and Avaya. It can also be used for digital platforms, including Slack, Facebook Messenger, and Google. The Dialogflow Customer Experience Edition (CX) enables the advanced development of chatbots and voicebots. It contains a visual platform for building bots, tools for modularizing bots, support for advanced interactive voice response (IVR), as well as tools for versioning and collaboration. It has been optimized and scaled for use at an enterprise-level.

Document Artificial Intelligence (AI) is not the appropriate choice for this scenario. You would use Document AI for the classification and extraction of structured data from existing documents. This can enable you to find key insights and automate organizational practices. You can use Human-in-the-Loop AI for workflow systems and a user interface to verify the information extracted from documents using Document AI.

Media Translation application programming interface (API) is not the appropriate choice for this scenario. You would use Media Translation API for automatic translation of audio between languages. This can also be done in real-time, if required. You can use Media Translation API for programmatic translation of audio inside of your applications. Media Translation API is based on the open source Google Remote Procedure Call (gRPC) framework.

You would not use Visual Inspection API for this scenario. You would use Visual Inspection API for training and deploying AI models for automatic detection, classification, and localization of any anomalies that are identified in images. This helps increase the quality of production in various industries and enables development of advanced analytics.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Dialogflow](#)

Related Material

Question #24 of 60

Question ID: 1495179

Your company is looking to build a new e-commerce portal for its customers and wants to take the appropriate cloud adoption approach. Which would be a suitable option for this requirement?

- A) First migrate the applications and then rearchitect them on the cloud**

✖
- B)** Rearchitect applications first to make them ready for the cloud

✖
- C)** Create a new infrastructure and application entirely on the cloud

✔
- D)** Use the cloud to only modernize the infrastructure

✖

Explanation

You will create a new infrastructure and application entirely on the cloud. This approach is also called “Invent” or “Greenfield”. To make this happen, a company needs to have access to a wide range of development skills and strong sponsorship from executives. This approach features the creation of Cloud Centers of Excellence (CCEs) which are teams comprised of people from different backgrounds who perform experimentation and create documentation for new methods of scaling and working on the cloud with a set of key best practices.

First migrating the applications and then rearchitecting them on the cloud is not a suitable approach for this scenario. This approach is also called “Move first and then change” or “Lift and shift”. This approach is better suited for a company that wants to adopt a conservative approach to cloud adoption. This method features fewer disruptions to existing work processes and operations. In this approach, the changes for cloud adoption take more time and gain momentum gradually as new work methods are accepted by more people in the company.

Rearchitecting applications first to make them ready for the cloud is not a suitable approach in this scenario. This is also called “Change first before moving” or “Improve and move”. This approach is suitable for companies that take a forceful approach for cloud adoption and features the use of a DevOps and automation-based approach to work as opposed to the traditional waterfall delivery model.

Using the cloud to only modernize the infrastructure is not suitable for this scenario. This approach is also simply called “Move”. This approach is for companies that just want to modernize their infrastructure layer to avail of virtualized infrastructure for disaster recovery (DR) and big data analysis. Big data can include structured data from databases and unstructured data from Internet of Things (IoT) sensors, images, emails, and social media.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > Managing Change in the Cloud](#)

Related Material

Question #25 of 60

Question ID: 1495191

Your company has various multi-cloud and hybrid cloud environments that store its data. You need to find a way to populate this data lake with data from all of these sources. Which service should you use to perform data ingestion for the data lake?

- A)** Dataproc

✖
- B)** Cloud Storage

✖
- C) Cloud Data Fusion**

✔
- D)** Cloud Trace

✖

Explanation

You should use Cloud Data Fusion for the process of data ingestion into the data lake. Cloud Data Fusion is a fully managed service that provides scalable integration for cloud-native data. It allows users to deploy Extract, Transform, Load/Extract, Load, Transform (ETL/ELT) data without requiring coding by using a visual interface. Cloud Data Fusion is created using the Centralized Data Analysis Platform (CDAP) data analytics platform, which is open source. This allows for the portability of data pipelines and integration for multi-cloud and hybrid cloud deployments that enable users to remove existing solos and access previously inaccessible data. Additionally, Cloud Data Fusion provides several preset transformations that can be used for stream and batch processing tasks. This boosts collaboration and productivity by allowing connections and transformations to be shared among users.

You would not use Cloud Storage for data ingestion. This is because it is an object storage service and not a data ingestion service. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria like a period having passed or a new version of the data created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

You would not use Dataproc for data ingestion. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, ETL operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use Cloud Trace for data ingestion. Cloud Trace is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Cloud Data Fusion](#)

Related Material

Question #26 of 60

Question ID: 1497128

What will an organization need to do to reach the transformational phase in its AI adoption journey?

- A)** Outsource complex ML problems and create a proof of concept using EDA tools and read-for-use AI tools
- B) Deploy customized ML models into production for addressing different use cases**
- C)** Build and deploy ML models from a single platform with cross-organizational accessibility
- D)** Create a Centre of Excellence (COE) containing data scientists for creating ML solutions for specific scenarios

Explanation

A transformational phase in artificial intelligence (AI) adoption is achieved when a company builds and deploys machine learning (ML) models from a single platform with cross-organizational accessibility, and implements an analytics team to create ML solutions for specific scenarios across various business functions. This phase is characterized by AI teams and data scientists that are specific to products working inside of existing product teams and creating ML models that are product-specific. There is also a central analytics team that creates patterns and recommended practices for taking projects requiring ML forward. A common ML platform provides ML components, models, datasets, and components needed for creating AI solutions. The platform also provides computing that is serverless and scalable for data ingestion and batch processing.

Creating a Center of Excellence (COE) containing data scientists for creating ML solutions for specific scenarios and deploying customized ML models into production for addressing different use cases are not characteristics of the transformational phase. These are both features of the strategic phase where a centralized team seeks to create ML systems for specific product teams and allocate ML talent as per business priorities. The team also looks to identify which ML products can be purchased instead of building them, and the team assesses use cases for feasibility.

Outsourcing complex ML problems and creating a proof of concept using exploratory data analysis (EDA) tools and read-for-use AI tools is a feature of the tactical phase and not the transformational phase. The tactical phase is characterized by ML projects within the company that are not fully aligned with the organization’s goals. At this stage, an organization is only exploring the feasibility of using AI for short-term goals and prototyping. An example of this is the use of AI systems for detecting written text, or building customer segmentation by utilizing descriptive analytics. At this stage, complex analytical problems are outsourced and ML projects within the company are done by individuals.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Services.google.com > Google Cloud's AI Adoption Framework](#)

Related Material

Question #27 of 60

Question ID: 1495901

You need to suggest a scaling strategy for a company that wants to move from its maturity level of strategic cloud adoption to one that is transformational. Which of these actions will you suggest for this requirement?

- A)** Keep the system immutable with no modifications
- B)** Perform each modification when issues are found
- C)** Perform each modification using a manual approach
- D) Perform each modification using a programmatic approach**

Explanation

You will suggest that the company perform each modification using a programmatic approach. This is because taking a transformational approach involves the use of automation. Automation ensures that changes are made in a constant and consistent manner with low risks to the system. This also ensures that changes can be rolled back, if necessary, to maintain a safe system configuration.

You would not perform each modification using a manual approach. This is not a transformational approach, but a tactical approach that involves changes being made infrequently and can pose high risks for the company. This approach is not recommended.

You would not perform each modification when issues are found. This is not a transformational approach that ensures that risks and problems are mitigated before they actually occur.

You would not keep the system immutable with no modifications. This is because a transformational approach requires rethinking and redefining existing systems and relationships for leveraging new technologies to facilitate operational excellence and business growth.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why it is critical for businesses to adopt new technology

References:
[Services.google.com > The Google Cloud Adoption Framework](#)

Related Material

Question #28 of 60

Question ID: 1497157

Your company's accounting department has raised an issue regarding the high operating costs of the cloud deployment. You are asked to find ways to optimize the costs of running the cloud. What should you do to reduce the Total Cost of Ownership (TCO) using Google Cloud?

- A) Utilize historical usage data for monitoring

✖
- B) Delete idle VM instances

✔
- C) Use larger VM sizes than needed

✖
- D) Store data on the most available storage classes

✖

Explanation

You should reduce the Total Cost of Ownership (TCO) through cost optimization tactics, such as deleting idle virtual machine (VM) instances. Cost optimization involves using managed services, keeping a consumption model, and analyzing expenditure and attributing it correctly. Cloud monitoring dashboards display key system metrics graphically in real-time, offering an updated view of the cloud system. This helps you get a clear analysis of how your systems infrastructure, applications, and services are doing. This way, you can use metrics and logs to track and resolve issues swiftly, thus improving your TCO and reducing Mean Time to Recovery (MTTR). You can also create alarms based on metrics. An example is ensuring that server sizes are selected to match actual computing needs in order to ensure greater cost-effectiveness.

You would not reduce the TCO by using larger VM sizes than needed. Using larger server sizes than required will cause a waste of resources and thus a waste of capital. Instead, you can opt for rightsizing VM instances based on your specific requirements. For this, you can create a custom machine type that allows you to specify the exact CPU and RAM configuration for your specific processing needs.

You would not reduce the TCO by using historical usage data for monitoring. Historical data cannot give you an updated view of a cloud system. Monitoring involves observing important system metrics in real-time using dashboards. This allows you to find ways to optimize costs through more efficient use of cloud resources. An example is monitoring VM instances to identify resources that may be idle.

You would not reduce the TCO by storing data on the most available storage classes. Using the most available storage class is suitable for high availability applications. However, it is also the most expensive, and not all types of data require high availability. By using more suitable storage classes for specific types of data, you can optimize storage costs and save on TCO. Google Cloud Storage offers various storage classes, which include standard, nearline, coldline, and archival. By moving data to less expensive storage classes, such as the archival class, you can save on costs for storing data that is accessed rarely. By using a lifecycle policy, you can automatically ensure that data objects change their storage class based on conditions you specify, which can be the age of the object.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > Best practices for optimizing your cloud costs](#)

Related Material

Question #29 of 60

Question ID: 1497125

You work for a data sciences company that is exploring cloud-based services for modernizing its data processing systems. You need to find a secure, fully managed service for managing clusters with Google Compute and Kubernetes systems. Which service will you use?

- A) Dataproc

✔
- B) BigQuery

✖
- C) Dataflow

✖
- D) Dataprep

✖

Explanation

You will use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence, and geospatial analytics ready for use. It has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. This enables you to perform queries on petabytes of data within minutes.

You would not use Dataprep for this scenario. Dataprep is a data service for exploring, cleaning, and preparing data visually for the purpose of ML and analysis. It is an intelligent data service and is serverless. This removes the need for users to deploy or manage infrastructure. It also suggests data transformations that are ideal each time a user interface (UI) input is detected, and this removes the need for a user to write code for these applications. Dataprep also detects any anomalies that may exist in the data, including duplicates and missing value data. This helps a user avoid having to assess the quality of data and proceed directly to exploratory data analysis.

You would not use Dataflow for this scenario. Dataflow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for data management

References:
[Cloud.google.com > Dataproc](#)

Related Material

Question #30 of 60

Question ID: 1495221

A game development company has been developing games from a legacy on-premises data center and is now considering making a digital transformation. How would the company benefit from modernizing its infrastructure with cloud technology?

- A) Delegating application security to the cloud service provider

✖
- B) Massive scalability by using serverless technology

✔
- C) Ensuring all resources are paid for regardless of usage

✖
- D) Making significant investments in new emergent technologies

✖

Explanation

Using cloud-based technologies like serverless computing, a company can achieve massive levels of scalability, which may not be feasible using traditional legacy systems. For example, BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

Delegating application security to the cloud service provider (CSP) is not a benefit of using cloud technology. Security of applications and all the data stored in the cloud is the responsibility of the customer and not the CSP. The CSP is responsible for the security of its data centers and of the cloud.

Paying for all resources, regardless of usage, is not a benefit of using cloud technology. Cloud computing ensures that companies only pay for resources consumed. This ensures that data centers do not need to be provisioned with more capacity to handle spikes in usage or organizational growth, and allows a company to have its staff work on more strategic tasks that help grow the business.

Making significant investments in new emergent technologies is not a benefit of using cloud technology. CSPs have access to the latest technologies, which customers can avail of as services. This removes the risk of the company having to make large investments into new technologies, which may become obsolete in time.

Some distinct benefits of cloud technology are outlined below:

- Security – Through a shared responsibility model and best practices, you can ensure greater safety of your operations using Google Cloud.
- Reliability – This helps you ensure your workloads can perform correctly, handle failures, and automatically fix any issues that may arise. This also covers data backups, fault isolation, and disaster scenario preparation.
- High Availability – This ensures that your applications are always ready to serve customers swiftly with almost no downtime. A measure of availability is five 9s, which corresponds to a system being available 99.999% of the time.
- Elasticity – This is the ability to scale your computing resources to match consumer demand without having to perform resource provisioning manually or guess future capacity needs.
- Agility – This is the ability to swiftly deploy new applications by removing unnecessary operational overhead so that the development team can focus on creating and innovating products.
- Pay-as-you-go (PAYG) pricing – This is a flexible payment option where a cloud consumer only pays for resources consumed.
- Scalability – This is using the right technology to easily make your application meet growing demands on a massive scale using serverless technologies.
- Global Reach – Using edge locations and data centers around the world, Google Cloud’s global infrastructure can improve user experience with low latency for end users, regardless of their location on the globe.
- Economy of scale – This is having lower PAYG pricing due to accumulated cloud usage from a very large number of cloud customers.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > What is Cloud Computing? > What are the benefits of cloud computing?](#)

Related Material

Question #31 of 60

Question ID: 1495193

A company that is in the digital publishing domain has approached you to help them find a suitable technology for building a data warehouse that they can use for performing analysis and reporting with scalability that is seamless. Which Google Cloud service should you recommend for this requirement?

- A) Cloud Trace

✖
- B) Cloud Storage

✖
- C) DataFlow

✖
- D) BigQuery

✔

Explanation

You should recommend BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use and has a serverless architecture allowing you to use SQL queries for gaining key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. This allows you to perform queries on petabytes of data within minutes.

You would not recommend DataFlow for this scenario because it is not a data warehousing service but a service for performing data processing. DataFlow is a fully managed service that automates the process of provisioning and managing processing resources. It provides the highest possible resource utilization by performing horizontal autoscaling for various worker resources.

You would not recommend Cloud Storage for this scenario because it is an object storage service and not a data warehousing service. Cloud Storage can be used for Object Lifecycle Management (OLM), where data can be configured so that data is automatically moved to lower-cost storage classes when the data meets specified criteria like a period having passed or a new version of the data created. Cloud Storage also provides data redundancy through various locations for storage buckets. This provides fast response times and disaster recovery (DR) options.

You would not recommend Cloud Trace in this scenario as it is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > What is BigQuery?](#)

Related Material

Question #32 of 60

Question ID: 1497124

Which of these are characterized as key advantages of moving to a data-driven culture? (Choose two.)

- A) Remove the need for data governance

✖
- B) Less reliance on fully managed services for data management

✖
- C) Use of a Zero Trust security model for data sharing

✔
- D) Easy access to analytics data for the business users of a company

✔
- E) Creation of duplicate data for use within an organization

✖

Explanation

Easy access to analytics data for the business users of a company and the use of a Zero Trust security model for data sharing are both key advantages of moving to a data-driven culture. A data-driven culture gives data the topmost priority and forms the basis of its decision making for its strategies and teams. A data-driven culture incorporates people, processes, and technology, and ensures that a business:

- operates with trust
- ensures insights are democratized
- increases its agility
- applies business intelligence

Less reliance on fully managed services for data management is not an advantage of moving to a data-driven culture. This is because data can be used efficiently when using more managed integrated systems. This bridges any skill gaps that may exist for users and enables them to operate, scale, and manage data with high levels of agility. This allows for greater user collaboration and removes bottlenecks and silos in an organization.

The creation of duplicate data for use within an organization is not an advantage of moving to a data-driven culture. A data-drive culture seeks to create a single source of reliable data within an organization that can be accessed by users when they need it. This can be achieved by ingestion of data into a data warehouse and building data models with business intelligence systems, such as Looker. This makes relevant data available for all teams without issues related to redundancy and provisioning overhead.

Removing the need for data governance is not an advantage of moving to a data-driven culture. Data governance is an integral part of a data-driven culture and ensures that only the correct users can access relevant data and also ensures that data is safe throughout its lifecycle.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Describe the role of data in digital transformation and the importance of a data-driven culture

References:
[Cloud.google.com > Google's guide to building a data-driven culture](#)

Related Material

Question #33 of 60

Question ID: 1495927

Your company wants to digitally transform its existing systems. Which of these are key business drivers for modernizing applications for your company? (Choose two.)

- A)** Remove the need for data gathering from changing markets
- B)** Increase staff productivity through greater change management
- C) Increasing system visibility and control over spread-out operations**
- D) Creation of more agile systems at optimized costs**
- E)** Increasing the TCO for facilitating necessary growth

Explanation

Increasing system visibility and control over spread-out operations and the creation of more agile systems at optimized costs are both key business drivers for modernizing applications. By modernizing applications and using cloud-native services, you can have granular control over your development and operational systems, which can be automated to ensure a robust, change-friendly system. With fully managed services and serverless technology, you can have more agile systems in place with lower costs due to lessened overhead for the company.

Increasing the total cost of ownership (TCO) for facilitating necessary growth is not a key business driver. The TCO would be significantly reduced when performing application modernization. This is because application modernization will require less investment by the company for leveraging the latest trending technologies as well as the ability to use managed services for greater scalability and agility. Application modernization ensures that less time and resources are spent on tasks like resource provisioning, scaling, backups, and version updating, which cuts down on the capital that would be required to implement similar functionalities on a legacy system.

Removing the need for data gathering from changing markets is not a key business driver. When modernizing applications, you are able to gather more data from both historical sources as well as trending markets using modern data gathering systems and tools for business intelligence (BI). Application modernization will enable you to use fully managed services for creating data lakes and data warehouses for big data analysis and use query services to deliver useful insights on the same for powering business decisions.

Increasing staff productivity through greater change management is not a key business driver. By using managed services and automation, you can ensure that staff members of varying technical skill levels can achieve greater productivity without the need for intensive change management. Managed artificial intelligence (AI) systems can ensure that even people without the necessary coding skills for creating machine learning systems can use cloud services to build systems that deliver business value by leveraging AI technology to better understand markets and customers.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Related Material

Question #34 of 60

Question ID: 1497146

Your company provides recruitment consultant services for a wide user base and needs a technology solution for classifying and extracting structured data from inside user application forms.

Which of these technologies would you use for this requirement?

- A)** Cloud Translation
- B)** Visual Inspection AI
- C)** Media Translation API
- D) Document AI**

Explanation

Document Artificial Intelligence (AI) is the appropriate choice for this scenario. You would use Document AI for classification and extraction of structured data from existing documents. This can enable you to find key insights and perform automation of organizational practices. You can use Human-in-the-Loop AI for workflow systems and a user interface to verify the information that is extracted from documents using Document AI.

Media Translation application programming interface (API) is not the appropriate choice for this scenario. You would use Media Translation API for the automatic translation of audio between languages. This can also be done in real-time, if required. You can use Media Translation API for programmatic translation of audio inside of your applications. Media Translation API is based on the open source Google Remote Procedure Call (gPRC) framework.

Visual Inspection AI is not the appropriate choice for this scenario. You would use Visual Inspection AI for training and deploying AI models for automatic detection, classification, and localization of any anomalies that are identified in images. This helps increase the quality of production in various industries and enables the development of advanced analytics.

Cloud Translation is not the appropriate choice for this scenario. Cloud Translation or Translation AI performs the translation of text between languages using Representational State Transfer (RESTful) APIs. You can use Cloud Translation to translate text from English to Spanish to any other language or vice versa. By using Cloud Translation API, you can provide automatic text translation features for an app or Website.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Document AI](#)

Related Material

Question #35 of 60

Question ID: 1495195

A company has an on-premises PostgreSQL server database system and is looking for a cloud-based alternative that is fully managed so the company can remove overhead costs associated with replication, maintenance, and updates. Which of these solutions should you recommend to meet this requirement?

- A) Dataproc**
- B) Cloud Data Fusion**
- C) Cloud Trace**
- D) Cloud SQL**
- 
- 
- 
- 

Explanation

You should recommend Cloud SQL for this scenario. Cloud SQL provides a relational database system that is fully managed and works with SQL Server, PostgreSQL, and MySQL. It provides features like flags for configurations, rich extension collections, and ecosystems for developers. Cloud SQL brings down the maintenance costs of databases and provides reliable operations supported by a site reliability engineering (SRE) team that is available around the clock. It has automation built-in for tasks like managing storage capacity and provisioning for databases. It can also be integrated with Google Kubernetes Engine (GKE) and BigQuery while also providing observability of your data using Cloud SQL Insights.

You would not recommend Cloud Trace in this scenario because it is used as a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights of its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

You would not recommend Cloud Data Fusion in this scenario because it is a fully managed service that provides scalable integration for cloud-native data. It allows users to deploy Extract, Transform, and Load/Extract, Load, and Transform (ETL/ELT) data without requiring coding by using a visual interface. Cloud Data Fusion is created using the Centralized Data Analysis Platform (CDAP) data analytics platform which is open source.

You would not recommend Dataproc in this scenario because it is better suited for applications related to data lakes. It provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, perform ETL operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Cloud SQL](#)
[Cloud.google.com > What is Cloud SQL?](#)

Related Material

Question #36 of 60

Question ID: 1495947

Your company has a photo sharing service that is heavily in use. You have created an SLO for the service. You need to find a way to check the status of your SLO and also have the error budget computed automatically. Which Google Cloud service would be suitable for this requirement?

- A) Cloud Monitoring**
- B) Looker**
- C) Cloud Vision**
- D) BigQuery**
- 
- 
- 
- 

Explanation

Cloud Monitoring can be used to create and use service level objectives (SLOs), service level indicators (SLIs), and error budgets. You can specify an SLO for a month, and Cloud Monitoring computes the corresponding error budget automatically. It can be used to monitor the complete infrastructure of a cloud deployment. It provides a view into the uptime, health, and performance of cloud systems. For this, Cloud Monitoring gathers relevant metadata, events, and metrics from sources that include uptime probes, services, app instrumentation, alerts, notifications, and other components of applications. It allows you to stop issues from happening, and it allows you to identify trends. It also reduces overhead related to monitoring through a service that is integrated and provides metrics, dashboards, and alerts all in one place. Cloud Monitoring provides SLO monitoring and metrics collection for virtual machines (VMs) (using Ops Agent) and Kubernetes (using Managed Service for Prometheus).

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence, and geospatial analytics ready for use, and has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure. This allows you to perform queries on petabytes of data within minutes.

You would not use Looker for this scenario. Looker is a platform for business intelligence that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications to provide embedded analytics. Looker can be used to create alerts for proactively sending timely insights related to various business data. It also has dashboards that are interactive, which allow for analyzing data in real-time through a trusted data model.

You would not use Cloud Vision for this scenario. Cloud Vision is used by developers to understand image content through machine learning (ML) models accessed through an application programming interface (API). It provides image classification into various categories, object and face detection inside images, and identification of printed words inside of images.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Cloud.google.com > Cloud Monitoring](#)

Related Material

Question #37 of 60

Question ID: 1497136

You need to present a case for adopting Google AI technology for driving your company's business innovation. Which of these actions will prove the value of using Google's Responsible AI for the company's business?

- A) Deploying ML models at scale without the overhead of model inspections

✖
- B) Reducing AI adoption and use for various enterprise systems

✖
- C) Relieving AI developers of making ethical considerations while designing systems

✖
- D) Creating products that have more safety and accountability

✔

Explanation

Creating products that have more safety and accountability is an example of how using Google's Responsible artificial intelligence (AI) is beneficial for a business. Creating accountability involves performing a detailed evaluation of AI products in scenarios where they perform as per expectations as well as when there are performance issues. This approach ensures that AI products can have accountability built into them.

Reducing AI adoption and use for various enterprise systems is not an example of using Google's Responsible AI. Using Responsible AI fosters an environment of trust leading to more enterprises and companies adopting AI systems. The underlying practices and commitments of AI development are a critical factor for companies when deciding to incorporate AI technology into their work.

Relieving AI developers of making ethical considerations while designing systems is not an example of using Google's Responsible AI. Ethical considerations are an integral part of a development process and needs to be accounted for. Responsible AI ensures that all developers and people making decisions while designing AI systems need to consider ethics when making design decisions. This ensures systems have greater transparency, responsibility, and adherence to ethics in their operation.

Deploying ML models at scale without the overhead of model inspections is not an example of using Google's Responsible AI. Model inspections need to be performed to ensure quality and performance of AI products. Responsible AI tools allow you to perform inspection and analysis of existing AI models to ensure safer products, increased customer trust, and ethically sound innovation. Model cards, TensorFlow open-source toolkit, and Explainable artificial intelligence (AI) are tools that are used as a part of Google's Responsible AI framework.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > Responsible AI](#)

Related Material

Question #38 of 60

Question ID: 1497159

You work for a multinational company that has recently performed a cloud adoption process. The company needs you to find ways to have more transparency into cloud expenses for better management. What cost management strategy should you use in this scenario?

- A) Identify any cloud cost overruns to the company's error budget

✖
- B) Delegate responsibility of cloud budgeting to one stakeholder

✖
- C) Facilitate collaboration between financial, IT, and marketing teams

✔
- D) Increase scrutiny of on-premises technology and services

✖

Explanation

You should facilitate collaboration between financial, IT, and marketing teams. This is necessary to ensure visibility and transparency as part of cost management. By incorporating a partnership that is implemented across various teams, you will gain greater visibility, which is necessary for efficient cost management. Reducing budget overruns and unauthorized resource use can be done with more collaboration between IT and finance teams.

You would not delegate the responsibility of cloud budgeting to one stakeholder. Keeping a single person in charge of managing cloud costs cannot work for a large organization due the scale and complexity involved. An individual can look after cloud cost management only for very small companies. Large companies will require cross-team collaboration for effective cost management.

You would not identify any cloud cost overruns to the company’s error budget. An error budget is not related to cloud expenses. It is the variation between the service level agreement (SLA) and service level objective (SLO). The error budget is the measure of the system’s possible downtime, which is the difference between the SLO and the total availability figure of 100%. The SLO is kept at a higher value than the SLA to ensure early warnings of possible availability issues.

You would not increase scrutiny of on-premises technology and services. The technology and services operating inside on-premises systems do not have an effect on the costs of running a cloud deployment. The financial management of on-premises systems is not related to the financial governance systems that need to be implemented to manage costs on a cloud deployment.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #39 of 60

Question ID: 1495233

Your company develops business software for various global markets. You need to ensure that your CI/CD pipeline is secure and any unsafe images are blocked from deployment. Which Google Cloud service should you use for this?

- A) Cloud Build

B) Dataproc

C) Firestore

D) BigQuery
- ✔

✖

✖

✖

Explanation

You should use Cloud Build for this scenario. Cloud Build is used to build, test, and deploy systems using a continuous integration and continuous delivery (CI/CD) platform, which is serverless. It allows a company to integrate its existing legacy systems and tools into a modern build system. This is made possible using third-party applications and customized steps for builds. Cloud Build also has vulnerability scanning features for CI/CD that prevents the deployment of any images that are unsafe. Some key features of Cloud Build include:

- Fast build time using VMs with fast CPUs provisioned to cache source code and dependencies.
- Deployment automation using pipelines with integrated use of Firebase, Google Kubernetes Engine (GKE), Cloud Functions, and App Engine.
- Multicloud deployment using multilanguage support for builder images used and containerized tasks, which are portable across clouds.
- Strong privacy through Google Cloud security, which allows the use of repositories for private source code inside private networks.

You would not use BigQuery for this scenario. BigQuery is a Google service that provides an enterprise-level data warehouse that is fully managed and that can be utilized for analytics. It has features like machine learning (ML), business intelligence (BI), and geospatial analytics ready for use. It also has a serverless architecture that allows you to use SQL queries to gain key organizational data. BigQuery has a scalable analysis engine that has a distributed structure and allows you to perform queries on petabytes of data within minutes.

You would not use Firestore for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database that is scalable.

You would not use Dataproc for this scenario. Dataproc provides a fully managed service that is scalable and allows you to run Presto, Apache Flink, and Apache Spark frameworks. You can perform data lake modernization, Extract, Transform, and Load (ETL) operations, and provide security for data science using Dataproc. It uses Apache Ranger and Kerberos for configuring security.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Cloud Build](#)

Related Material

Question #40 of 60

Question ID: 1495933

Your company provides various services to users online through endpoints. You want to find a fully managed service to make your system scalable and usable for your team members who don’t have strong coding skills, while also ensuring high security of the published applications. Which of these Google Services should you use for this requirement?

- A) Looker

B) Payment Gateway

C) API Gateway

D) GKE
- ✖

✖

✔

✖

Explanation

You should use API Gateway for this requirement. API Gateway provides a gateway that is fully managed for developing, managing, deploying, and securing APIs for serverless backend systems. These backend systems can include App Engine, Cloud Run, and Cloud Functions. API Gateway is cost optimized with billing done based on tiers and consumption. It is built on Envoy, which is a communication system for modern service-based architectures that includes edge and service proxies for cloud-native systems. API Gateway provides the flexibility of deployment, superior performance, scalability, and ease of use, thus allowing teams to focus on developing effective applications. API Gateway also provides ways to monitor, trace, log APIs, and create alerts. It has systems built-in for the validation of keys and authentication to protect your online services.

You would not use Looker for this scenario. Looker is a platform for business intelligence (BI) that allows you to access intelligent data for making better business decisions. It also can be embedded into other applications to provide embedded analytics. Looker can be used to create alerts to proactively send timely insights related to various business data. It also has dashboards that are interactive, which allow you to analyze data in real-time through a trusted data model.

You would not use Payment Gateway for this scenario. With Payment Gateway, you can perform processing of payments that are application programming interface (API)-based. It can scale up to cater to unpredictable surges in usage as well as be extended to third-party insights and anomaly finding tools through APIs. It also allows for payment processing across multiple regions of Google Cloud's global infrastructure. Payment Gateway also has built-in tools that allow you to monitor, debug, and trace API-based purchases with high availability and less overhead. You can also export data of transactions into a project on Google Cloud using an export API.

You would not use Google Kubernetes Engine (GKE) for this scenario. GKE is not used for API management, which this scenario needs. GKE allows you to create and manage Kubernetes clusters. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications. GKE provides detailed control over containers and nodes, enabling you to specify CPU, RAM, storage, GPUs, and network configurations along with an Autopilot option for situations where there is not enough operational and developer support available.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Cloud.google.com > API Gateway](#)

Related Material

Question #41 of 60

Question ID: 1497149

A bank has an on-premises data center that it needs to maintain to ensure regulatory compliance. It also has several applications that it uses for internal purposes as well as for serving its customers. What should you use to provide a single portal for visibility of their nodes and clusters in their cloud and on-premises workloads that also manages large volumes of Kubernetes clusters inside their applications?

- A) Anthos

B) App Engine

C) Firestore

D) Cloud Functions
- ✔

✖

✖

✖

Explanation

You should use Anthos for this scenario. Using Anthos, you can create and manage an application and then run it on both on-premises systems and on the cloud. With Anthos, you can perform migrations straight from virtual machines (VMs), and create and deploy apps using Google Kubernetes Engine (GKE), VMs, and landing zones that are serverless. Anthos allows you to use its platform for deploying both on-premises and cloud-native applications to create a global fleet that has consistency in its operations. Using containers and serverless services, Anthos lets you grow productivity. By maintaining security systems at every stage of the application's lifecycle, Anthos provides protection for both a software supply chain and application.

You would not use App Engine for this scenario. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine, you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning. App Engine provides support for web application development using programming languages like Python, Ruby, C#, Node.js, Go, and PHP.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources, such as Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). For example: running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not use Firestore for this scenario. Firestore is a serverless document database that is scalable and fully managed. It enables fast development for applications for the internet, mobile, and the Internet of Things (IoT), which can connect to the database directly. Firestore can be used to create real-time applications by using its native synchronization features. It also allows you to customize various rules for security and data validation to protect your data. Firestore can be integrated with BigQuery and Firebase. Firestore is an upgrade to the older service called Datastore, which provides a NoSQL database that is scalable.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Anthos](#)

Related Material

Question #42 of 60

Question ID: 1495281

Your company has started an initiative to upgrade its existing video streaming service for its customers. The company needs to ensure that its SLA for the service to its customers is met. You have an SLA of 99% defined for service availability. What should be the value of your SLO?

- A) 89%

B) 99%

C) 99.5%

D) 98.99%
- ✖

✖

✔

✖

Explanation

You should use the value of 99.5% for the service level objective (SLO). This is because you need to keep an SLO value that is higher than the service level agreement (SLA), which helps you get early warnings when there is a danger of not meeting the SLA. Failure to meet a set SLA can incur financial penalties for the company. SLAs are visible to customers that use a service while SLOs and SLIs are internal to the development and operations team. If you do not meet an SLO, then there is a possibility of not meeting the SLA, which can cause issues for the company.

The SLO of 99% is incorrect because this value does not allow you to have an early warning of SLAs not being met. You need to keep the SLO value higher than the SLA, as a rule. The SLO of 99% is the same as the SLA of 99%, which indicates that failure to meet the SLO will mean failure to meet the SLA and indicates that there is no early warning possible for the SLA not being met. You need to specify an SLO that exceeds the SLA. An SLO specifies a definite target for a service, which can be 99.999% uptime. SLOs are defined based on user experience and can address user-centric needs, such as latency.

The SLO of 89% and 98.99% are both incorrect because these SLO values are lower than the SLA value of 99%. This raises an unacceptable situation where SLOs will not meet the SLA. For this reason, you need to ensure that the SLO value exceeds the SLA to create a system where there is no possibility of the SLA not being met.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #43 of 60

Question ID: 1495287

A company has an SLA in place for its service and wants to ensure that it is met. What can be the potential impact of unexpected downtime on the system? (Choose two.)

- A) Addition of new features

✖
- B) Financial losses

✔
- C) Greater error budget

✖
- D) SLA not being met

✔
- E) Increased ROI

✖

Explanation

System downtime can lead to service level agreements (SLAs) not being met, which can then incur financial losses for a company. SLAs are an agreement between the service provider and customers that specifies the level of availability that the service will have. To ensure that an SLA is met, you need to keep a service level objective (SLO) value that is higher than the SLA. This helps you get early warnings when there is a danger of not meeting the SLA. SLAs are visible to customers that use a service while SLOs and SLIs are internal to the development and operations team. If you do not meet an SLO, then there is a possibility of not meeting the SLA, which can cause issues for the company.

The addition of new features is not a potential impact of downtime. During system downtime, there cannot be new features added. Developers may want to add new features to the system during application development, while operations teams want to minimize any possible disruptions to the service and seek to minimize any modifications to the service. This creates a conflict between the teams. Creating a common SLO ensures that there can be an agreement between operations and development teams to meet a common goal of ensuring service uptime while adding new features.

An increased return on investment (ROI) is not a potential impact of downtime. The ROI for a company can be adversely affected by prolonged or unexpected downtimes, as this can cause customers to leave. Monitoring costs and measuring ROI are key functions of business teams and management and are important to ensure that the organization's financial performance is in line with the strategy and cost management principles of the company.

A greater error budget is not a potential impact of downtime. The occurrence of prolonged system downtime indicates that the error budget is exhausted. The error budget is the measure of the system's possible downtime, which is the difference between the SLO and the total availability figure of 100%. When an SLO is agreed upon, its corresponding service downtime or error budget is also agreed upon. An SLO specifies a definite target for a service, which can be 99.999% uptime. SLOs are defined based on user experience and can address user-centric needs, such as latency.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Identify Google Cloud solutions for cloud resource monitoring and application performance management

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #44 of 60

Question ID: 1495273

You work for a government agency that has sensitive data stored on its hybrid cloud environment that is accessed by authorized users across the globe. You need to implement a security solution that can use machine learning to protect your system from DDoS attacks. Which of these services should you use for this?

- A) Cloud Endpoints

✖
- B) Cloud Armor

✔
- C) Cloud Trace

✖
- D) Cloud Spanner

✖

Explanation

You should use Cloud Armor for this scenario. Cloud Armor is a service that protects websites and software applications from threats like web attacks and distributed denial of service (DDoS) attacks. It offers adaptive protection by detecting and mitigating large numbers of Layer 7 DDoS incidents using a locally trained machine learning (ML) system. Cloud Armor has support for both cloud and hybrid cloud environments and contains web application firewall (WAF) rules that are preset for protection against common web application threats and Open Web Application Security Project (OWASP) Top 10 cyberthreats. It also has integration with reCAPTCHA Enterprise for providing protection against bot attacks as well as rate limiting features to protect your application from illegal requests that may be flooding your instances.

You would not use Cloud Trace in this scenario. Cloud Trace is a system for tracing that gathers data related to latency from various applications. The Cloud Trace service is useful for identifying any bottlenecks that may exist in your system. The latency data collected by Cloud Trace is displayed on the Google Cloud Console. This lets you track the propagation of requests in your application and view near real-time insights into its performance. Cloud Trace can be used to trace performance data from containers, virtual machines (VMs), and projects for App Engine. It also provides automatic alerts when it detects a deviation from the latency profile of your app.

You would not use Cloud Endpoints in this scenario. Cloud Endpoints provide a system for managing application programming interfaces (APIs) that enables monitoring, analyzing, and securing APIs as well as specifying quotas for them. You can monitor the key operational metrics for the APIs using Google Cloud Console and gain usage insights with BigQuery, Cloud Trace, and Cloud Logging. You can also utilize Cloud Endpoints Portal to create a website that API users can access to find documentation for the APIs and interact with them.

You would not use Cloud Spanner in this scenario. Cloud Spanner is a relational database service that is fully managed and ideal for critical business applications. You would use Cloud Spanner for provisioning a highly scalable database for online transaction processing (OLTP). This database will have high levels of availability and consistency worldwide.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Google Cloud Armor](#)

Related Material

Question #45 of 60

Question ID: 1495188

An organization is building a modern, unified analytics platform using Google Cloud. It must decide between using a data lake or a data warehouse for its data analytics needs. Which of these would be an appropriate business use case for using a data warehouse?

- A) There is a need to explore insights using existing resources**
- B) Known datasets need to be analyzed and preset questions answered**
- C) There is no clarity on the kinds of analysis that may need to be done on the data**
- D) There is a need to discover data across various data types**

Explanation

When known datasets need to be analyzed and preset questions answered then a data warehouse would be the appropriate choice for an organization. A data warehouse is a massive repository that contains a company's data gathered from various different sources. This data includes current data as well as data that is historical. A data warehouse is used for business intelligence (BI) activities like complex analytics which can lead to making informed management decisions. The sources of data for a data warehouse include customer relationship management (CRM) systems, automation systems for marketing, applications for transactions, and log files from applications. A data warehouse can be used for performing analysis in an ad hoc manner or for creating custom reports.

The other options are all characteristics of business use cases that can benefit from using a data lake:

- There is a need to discover data across various data types
- There is a need to explore insights using existing resources
- There is no clarity on the kinds of analysis that may need to be done on the data

A data lake allows you to store both structured, semi-structured, and unstructured data from multiple sources in a single place. This data can come from a variety of sources, including multimedia, social media, hard disks, servers, Internet of Things (IoT) devices, clickstream, databases, and transactional information. A data lake provides security and scalability and enables companies to perform ingestion of data from various sources including edge computing, cloud, or on-premises systems. The data can be stored in its original format and processed as a batch or in real-time. This data can then be analyzed using tools like Python, SQL, and R.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Services.google.com > Build a modern, unified analytics data platform > Data warehouse or data lake?](#)

Related Material

Question #46 of 60

Question ID: 1497135

Your company is in the process of adopting Google AI technology for its business growth. You are evaluating fairness and security in Google's AI and machine learning differentiators. Which of these is a recommended best practice for Google's stipulated responsible AI?

- A) Perform engagement with multiple users in the development process**
- B) Conduct short-term fixes to all issues for swift problem solving**
- C) Keep all the features in your ML model, including redundant ones**
- D) Train models to create automatic inferences based on detected correlations**

Explanation

Performing engagement with multiple users in the development process is a recommended best practice for Google's stipulated responsible artificial intelligence (AI) framework. This is part of a best practice of utilizing a design approach that is centered on actual users. The effectiveness of an AI system's predictions and decisions can be gauged from the experiences of actual users. Incorporating inputs from a wide range of users and use cases helps enrich the project with a range of perspectives from users. This eventually benefits the users that actually utilize the technology. The system needs to have enough built-in disclosures to users so that a satisfying user experience can be achieved.

Keeping all the features in your machine learning (ML) model, including redundant ones, is not a recommended best practice for Google's stipulated responsible AI. The recommended practice is to use the ML model, which is simple and meets goals for performance without using redundant features. The raw data used by an ML model needs to be analyzed, and if the model uses sensitive data, then that data should be understood within privacy mandates for the data. The data needs to be checked for errors, bias, and training-serving skews to ensure the optimal performance of the model.

Training models to create automatic inferences based on detected correlations is not a recommended best practice for Google's stipulated responsible AI. You need to ensure that the ML model does not make casual inferences based on correlations that it detects. You also need to be aware of the limitations of the model based on its training data's scope and range. Communicating these limitations to users can help with getting useful feedback about the AI system from users.

Conducting short-term fixes to all issues for swift problem solving is not a recommended best practice for Google's stipulated responsible AI. Short-term fixes may save time but do not offer solutions that are beneficial in the long-term. For this reason, it is important to create a balance between using short-term fixes and long-term solutions. Also, you need to compare the candidate ML model with the deployed version to identify the ways they differ, and to understand how the update will alter the user experience and system performance of the AI application.

Some additional best practices include using metrics like user feedback, false negatives and positives, product health reports, and system performance data for assessing monitoring and training of the ML model. Also, testing the system with both integration tests as well as unit tests to ensure that the performance of the AI system is as expected. This involves studying how the individual components of the ML model work with other components of the application. Also, continuous monitoring of the model needs to be done to ensure the model incorporates feedback from the real-world and user experiences.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Ai.google/responsibilities > Responsible AI practices](#)

Related Material

Question #47 of 60

Question ID: 1495230

Your organization uses virtual machines for their main database processing application. Each time new capabilities need to be added to the application, the system needs to be brought offline, which causes significant downtime. What would you suggest as a more robust solution for this situation?

- A) Upgrade the CPUs for the underlying infrastructure

✖
- B) Use containers to compartmentalize the application

✔
- C) Use high performance solid-state drives for storing application data

✖
- D) Install updated hypervisors for running virtual machines

✖

Explanation

You should suggest using containers to compartmentalize the application in this scenario. By compartmentalizing an application using containers, you can then change parts of the application in isolation from the other parts. Containers are software packages that contain application code as well as dependencies, such as runtime environments and libraries, that are required for running an application. This allows containers to virtualize an OS and allows applications to be made portable across multiple systems. Google search, YouTube, and Gmail are all containerized applications. Containers allow you to:

- Separate responsibilities – This allows development teams to work on application logic and features and operations teams to focus on application deployment.
- Make workloads portable – This allows containers to be run from any OS like Windows, Mac, or Linux and on any system including virtual machines (VMs), physical servers, a developer's laptop, on-premises systems, or a public cloud.
- Isolate applications – Containers perform OS-level virtualization of resources like memory, CPU, storage media, and networking systems which isolates the OS from any other applications.

A key difference between containers and VMs is that containers perform virtualization at the level of the OS while VMs perform virtualization at the level of hardware.

You would not use high performance solid-state drives (SSDs) for storing application data. The storage medium for the system is not a contributing factor in performing application compartmentalization, which this scenario requires. You can use a variety of storage media for your system which can include hard disk drives (HDDs) or SSDs depending on your specific requirements.

You would not install updated hypervisors for running VMs. Hypervisors allow VMs to optimally execute several applications on VMs concurrently. However, they do not have provisions for compartmentalizing applications.

You would not upgrade the CPUs for the underlying infrastructure. Changing the CPUs can increase the processing capabilities of the underlying system but this will not alter the application architecture which needs to be compartmentalized using container technology.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > What are Containers?](#)

Related Material

Question #48 of 60

Question ID: 1495243

Your company runs several workloads using containerized applications with specific OS and networking configurations. You need to create and manage clusters with granular control at the nodal level and have a way of automatically running the clusters without monitoring.

How should you do this?

- A) GKE

✔
- B) Compute Engine

✖
- C) App Engine

✖
- D) Cloud Functions

✖

Explanation

You should use Google Kubernetes Engine (GKE) in this scenario. GKE provides a fully managed Kubernetes platform that has scalability and automation. It provides granular control over containers so you can specify the exact OS, memory, storage disk, CPU, and networking configuration you need. GKE has a fully managed Autopilot mode that manages the complete infrastructure of a cluster without the need for monitoring or configuration. This mode also allows for cost optimization by ensuring billing is done only for utilized pods and not for system overhead, components, or unused resources. The Standard mode of GKE allows you to implement more granular controls and have customized workloads for administrative tasks. Kubernetes provides an open-source system to automatically manage, deploy, and scale container-based applications.

You would not use App Engine for this scenario. App Engine allows you to create and host your applications on the same platforms Google uses for several of its own applications. With App Engine, you can build and deploy applications quickly and manage them easily. It provides scalability without the issues related to backups, patching, and hardware provisioning. App Engine provides support for web application development using programming languages like Python, Ruby, C#, Node.js, Go, and PHP.

You would not use Cloud Functions for this scenario. Cloud Functions is best suited for when you need to run functional code based on an event or a trigger. Cloud Functions runs your code when events occur from various sources like Pub/Sub, Cloud Storage, and Hypertext Transfer Protocol (HTTP). For example: running a process for video transcoding automatically whenever a video is uploaded to Cloud Storage.

You would not use Compute Engine for this scenario. Compute Engine provides VM computing that has scalability and flexibility. It allows you to utilize Cloud Tensor Processing Units (TPUs), Graphics Processing Units (GPUs), and CPUs to perform analytics and processing on a large scale using Google's technological infrastructure.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand modernizing applications with Google Cloud

References:
[Cloud.google.com > Google Kubernetes Engine](#)

Related Material

Question #49 of 60

Question ID: 1495283

You work for a software development company that makes two deployments a day for its workplace collaboration tool, which is heavily in use and growing daily. Your development team has come up with a lot of new features that it wants to release, but the specified error budget for the month has been consumed. How will you roll out future deployments?

- A) Once a week

✖
- B) Once a month

✖
- C) Halt deployments

✔
- D) Once a day

✖

Explanation

You will halt deployments. This is because when new features are added to a service, new bugs may be introduced, which can eat into an existing error budget. There can then come a stage where the error budget for a given period is exhausted, and management will need to halt development of new features until the error budget is restored. For example, when a company that deploys code changes twice a day consumes half its error budget, it needs to reduce code deployments to once a day. When three-fourths of the error budget is consumed, the deployments can be reduced to once a month. When the error budget hits zero for a month, deployments will need to be halted.

The other options are incorrect because you cannot roll out any new deployments when the error budget is consumed for the month. This can cause a failure to meet the service level agreement (SLA) with the service users and cause financial loss for the company. The error budget is the measure of the system's possible downtime, which is the difference between the service level objective (SLO) and the total availability figure of 100%. When an SLO is agreed upon, its corresponding service downtime or error budget is also agreed upon. An SLO specifies a definite target for a service, which can be 99.999% uptime. SLOs are defined based on user experience and can address user-centric needs, such as latency.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

Question #50 of 60

Question ID: 1497151

A company's cloud deployment has already had a lot of budget overruns, and the company needs to cut back on cloud expenditure immediately. You need to implement financial governance strategies to gain more control over cloud costs. How would you achieve this?

- A) Providing cost viewing to all cloud users

✖
- B) Using organizational policies for resources

✔
- C) Decommissioning unused on-premises servers

✖
- D) Ensuring each user can provision cloud resources

✖

Explanation

You would use organizational policies for resources to gain more control over your cloud costs. The purpose of using organizational policies is to specify permissions for users to perform spending on Google Cloud based on their role in the organizational hierarchy. This way, you control the ability of specific users to view costs and perform administrative functions using Cloud Identity and Access Management (Cloud IAM) for cloud resources. You will also use cost management controls to ensure you don't spend beyond your stipulated budget and plan costs properly. You can set limitations on cost accumulations using quotas and perform monitoring for usage and costs using budgets. You can perform automated throttling and capping of resource consumption as well as broadcast budget notification alerts on company communication channels, such as Slack.

Providing cost viewing to all cloud users is not the correct way to gain more control over your cloud costs because each user on a cloud deployment cannot be allowed more privileges than their job role requires, as this can cause security as well as cost management issues. Cloud resource usage and cost viewing permissions need to be made based on the specific role a user performs in the organization.

Ensuring each user can provision cloud resources is not the correct way to gain more control over your cloud costs because allowing all users to provision cloud resources can cause major cost overruns and cost control issues. Correct financial governance on the cloud is to allow resource provisioning based on the user's role and position in the organizational hierarchy.

Decommissioning unused on-premises servers is not the correct way to gain more control over your cloud costs because the state of on-premises infrastructure does not affect cloud operating costs as the two systems are disparate. Decommissioning unused servers from an on-premises data center can help improve the operational effectiveness of an on-premises data center; However, this will not have an effect on how resources are provisioned on the cloud deployment, which requires a different system be in place to ensure cost control, such as using IAM policies.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe financial governance in the cloud and Google Cloud's recommended best practices for effective cloud cost management

References:
[Cloud.google.com > A guide to financial governance in the cloud](#)

Related Material

Question #51 of 60

Question ID: 1497405

You want to ensure maximum security of your sensitive data and also leverage cloud-native services like serverless computing. Which cloud deployment should you opt for?

- A) Hybrid Cloud

B) Private Cloud

C) Multi-Cloud

D) Public Cloud
- ✔

✖

✖

✖

Explanation

You should opt for a hybrid cloud infrastructure model. In a hybrid cloud system, a company uses its own data center along with a public cloud system to achieve its business goals. A hybrid cloud always contains a private cloud. A hybrid cloud deployment allows a company to run its applications on-premises and use the public cloud for bursting when user demands go beyond the limits of the data center.

You would not opt for a private cloud infrastructure. In this deployment, all the infrastructure and applications that are part of the cloud or virtualization system are running on systems that are inside a company's on-premises data center. This deployment option does not include systems that are a part of a public cloud service offering.

You would not opt for a public cloud infrastructure. In this deployment, the infrastructure includes applications that are fully functional in a cloud and have no on-premises portions. This deployment option may not be suitable for situations where sensitive data needs to be stored locally and not on a public medium.

You would not opt for a multi-cloud infrastructure. In this deployment, the data is still stored on a public system, which can cause security concerns for the company. A multi-cloud deployment involves using multiple public CSPs where each public cloud provides a different service for the company. This allows a company to use one public cloud for database services and another one for analytics. This deployment option lets a company use a preferred cloud provider for specific services. Remember that using multiple public CSPs indicates a multi-cloud infrastructure.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Hybrid and multi-cloud architecture patterns](#)

Related Material

Question #52 of 60

Question ID: 1495899

You work for a software development company that creates application software for desktop and mobile users as well as custom designed operating systems. The company now needs to migrate to a cloud-based system that can provide all the physical hardware it needs. The company will still utilize its self-owned software and applications.

Which cloud service model would you use based on this requirement?

- A) SaaS

B) DaaS

C) PaaS

D) IaaS
- ✖

✖

✖

✔

Explanation

You would deploy the Infrastructure as a Service (IaaS) model. This provides the necessary hardware via the cloud that the company can then utilize to run its applications and even install a customized operating system (OS). The IaaS model allows the use of barebones computing resources, such as physical hardware and networking, so that consumers can install the OS of their choice and run any applications they wish. The advantage of this is that a company's existing data center hardware resources can be completely replaced with a cloud-based solution. The IaaS model provides networking, storage, processing, and other basic computing resources for consumers.

The other options are incorrect because they do not allow the installation and use of a customized OS and they are better suited for needs related to consumers who want an existing software environment.

For the exam, you need to know some key concepts related to cloud computing. Cloud computing is the provision of computational services made available to consumers by cloud providers based on their need or consumption. These services cover both the use of platforms and applications as well as physical hardware resources. There are three primary cloud service models that provide services accessible by Web browsers, cellphones, and similar devices. The three primary cloud service models are PaaS, SaaS, and IaaS.

The Platform as a Service (PaaS) model allows consumers to install their own applications onto a cloud platform. The provider supplies the necessary OS and physical hardware. This allows consumers to quickly deploy their own applications without the burden of first acquiring the necessary physical resources and OS to make it happen. The PaaS model provides compute services, OS, networking, storage, and all the necessary hardware.

The Software as a Service (SaaS) model covers the use of software applications by consumers who are running on a provider's cloud infrastructure. These applications can be accessed using thin client interfaces, including cellphones and Web browsers. An example of this is email accessible from a Web browser, such as Gmail. The advantage of this model is that a consumer can utilize the application without the need to physically install the application or be concerned about the infrastructure needed to run it. The cloud provider takes full responsibility for running the application and managing it. Other examples of utilities that fall under the SaaS model include Enterprise Resource Planning (ERP), software development programs, Customer Relationship Management (CRM) systems, and Human Resources (HR) applications. Software applications for storage, networking, and processing also comprise the SaaS model.

Other cloud service models include:

- Communications as a Service (CaaS) – This service model includes video conferencing, chat, voice calling, emailing, and other means of collaboration.
- Anything as a Service (XaaS) – This service model includes the provision of complete IT solutions as a consolidated unified package covering end-to-end services.
- Desktop as a Service (DaaS) – This service model provides a virtual desktop that can be accessed by a variety of devices, including laptops, tablets, and cellphones. The desktop has all of the applications typically required by business workflows, including word processors and spreadsheets.
- Business Process as a Service (BPaaS) – This service model provides applications supporting routine business operations, such as shipping, orders, inventory, and payroll.

Objective:
Introduction to Digital Transformation with Google Cloud

Sub-Objective:
Explain why cloud technology is revolutionizing business

References:
[Cloud.google.com > What is IaaS?](#)

Related Material

Question #53 of 60

Question ID: 1495189

A company currently uses its on-premises data center for data storage, and they have a varying number of users accessing their website each month. The company is considering moving to a cloud deployment. Which of these advantages can they expect to gain from this decision?

- A) Adoption of a capital expenditure system

✖
- B) Service availability at 100%

✖
- C) Scalability for storage that is agile

✔
- D) Source code debugged by the cloud service provider

✖

Explanation

The company can expect scalability for storage that is agile. This is a key advantage of using cloud technology as it ensures that the company will only need to pay for resources that they actually consume. Additionally, due to the scalability of the cloud, the company can avail of the benefit of elasticity, where resources can be automatically added or removed to match the fluctuation in the amount of website usage.

The company cannot expect service availability at 100%. This is because a realistic measure of service availability is the five 9s uptime, which corresponds to services available 99.999% of the time. This can indicate downtime of under 6 minutes per year. Agreements related to system uptime are done using a service level agreement (SLA) between a cloud service provider (CSP) and the customer. The SLA specifically provides a formal definition of uptime. An uptime measure of five 9s indicates a highly available system. As an example, the Google Compute Engine SLA specifies that instances in multiple zones will have a minimum monthly uptime of 99.99%. Google Cloud Storage offers 99.999999999% or eleven 9s of annual durability.

The company cannot expect source code debugged by the CSP. This is because even when a company's infrastructure moves from an on-premises data center to a cloud deployment, the company's development team will still be fully responsible for the source code. Cloud deployments follow a shared responsibility model where the CSP is responsible for the cloud infrastructure and its security, and the company is responsible for the data that is stored and used in the cloud.

The company cannot expect the adoption of a capital expenditure system. This is because when a company moves to a cloud deployment, it essentially moves from the existing capital expenditure system to an operational expenditure system. The key difference between the two is that in a capital expenditure system, the company will take on the costs of procuring all the infrastructure and resources they need for running its IT systems. In this system, underutilization of resources and scalability issues can lead to wasted capital for the company. To counter this, an operational expense model allows a company to pay for resources consumed for its operations dynamically. This is also characterized by the pay-as-you-go (PAYG) model, where a cloud customer only pays for resources consumed.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify common Google Cloud solutions for smart analytics

References:
[Cloud.google.com > Compute Engine Service Level Agreement \(SLA\)](#)

Related Material

Question #54 of 60

Question ID: 1495930

A company with an on-premises data center needs to understand the benefits of adopting an API-based system. Which of these tasks using APIs can modernize an existing legacy on-premises system?

- A) Splitting applications into individual business services

✔
- B) Ensuring all teams perform tasks at the same pace

✖
- C) Reducing automation to give developers more control

✖
- D) Creating a reliable monolithic application for serving users

✖

Explanation

By using application programming interfaces (APIs), you can split existing applications into individual business services. The process of functional decomposition involves identifying individual components and their related systems inside of existing workloads. Then, you can create a decoupled architecture where workloads are broken into smaller parts and scaled independently. This can help you provide higher availability of services and protection from failures. With APIs for various business services, multiple service users and teams can use these services at their own pace. This system also provides greater visibility into the usage of these services in the company.

Reducing automation to give developers more control is not a benefit of using APIs. One of the main advantages of using APIs is to provide automation that reduces the time users spend doing tasks by having more tasks performed by computers. This boosts productivity and also can provide more accountability with tracking systems that record API calls.

Creating a reliable monolithic application for serving users is not a benefit of using APIs. APIs help move away from a monolithic architecture for application development to a microservices-based architecture. This provides greater stability to the system and ensures higher availability as individual components of the system can run independently, and there is no dependency between them. This ensures that if one component fails, then the other application components are not affected.

Ensuring all teams perform tasks at the same pace is not a benefit of using APIs. APIs can be used for various business services, and multiple service users and teams can use these services at their own pace. This system also provides greater visibility into the usage of these services in the company.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Understand the value of APIs

References:
[Inthecloud.withgoogle.com > CIO Guide to Application Modernization](#)

Related Material

Question #55 of 60

Question ID: 1495941

Your company has various business services in use that require the procurement of third-party software. You need to do a security analysis of this approach. What can be the potential problems that can occur when using third-party software in an existing cloud deployment?

- A) Third-party software lacks the support systems available for cloud-based software

✖
- B) Third-party software cannot handle important business services

✖
- C) Third-party software reduces a company's innovation goals

✖
- D) Third-party software can be prone to cyberattacks

✔

Explanation

Third-party software can be prone to cyberattacks. Third-party software can have security issues in it, which can compromise an organization's data security and make the organization vulnerable to cyberattacks. There needs to be a comprehensive security architecture in place within a company's IT processes to check third-party software and monitor it on a regular basis to ensure there are no security issues.

Third-party software has ample support systems available through various channels that include vendor support, online forums, and support groups. Third-party software can be developed by companies or groups that have a strong grasp of cybersecurity challenges and take the necessary measures and perform development best practices to ensure the software is secure and error free. The security level of software depends greatly on the development methodology employed to create it as well as the level of security awareness of the team that created it. Some open-source frameworks have forums and collaborative groups where open knowledge sharing contributes to mitigating any existing issues.

Third-party software does not reduce a company's innovation goals. The level of innovation that a company can incorporate into its operation depends more on the scale to which it has embraced digital transformation and not on its reliance on third-party software. To reach a state of being fully innovative, an organization will first need to ensure that it has open knowledge sharing, decentralized teams, more autonomous decision making, and use of artificial intelligence (AI) systems by team members of varying technical skill.

Third-party software can handle important business services and serve various critical functions of an organization. The amount of capability available in third-party software varies across industries and vendors. Third-party software can be as powerful as cloud offerings and sometimes even exceed the current capabilities of a cloud software offering. You need to select the right third-party software for the task at hand and make the decision based on various factors, which include the security of the third-party software.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Google security overview](#)

Related Material

Question #56 of 60

Question ID: 1497406

Your operations and development teams need to create a plan for ensuring service availability while adding some key enhancements. For this purpose, which of these can you consider valid service level indicators (SLIs) to meet your service level objectives (SLOs)? (Choose two.)

- A) Amount of memory consumed by the service

✖
- B) CPU utilization for the service

✖
- C) Percentage of correct user responses

✔
- D) Number of API calls generated for the service

✖
- E) Number of successful login attempts

✔

Explanation

The percentage of correct user responses and the number of successful login attempts are both valid service level indicators (SLIs) that you can use to meet your service level objectives (SLOs). SLIs measure the quantity of a specific aspect of the service that is relevant to the user. SLOs are defined based on user experience and can address user-centric needs, such as latency. SLIs help gauge if SLOs are met. A combination of SLOs and SLIs work towards meeting service level agreements (SLAs), which specify the exact agreement made with the users of the service.

CPU utilization for the service, the amount of memory consumed by the service, and the number of application programming interface (API) calls generated for the service are all not the best options for SLIs as these are internal to the system and not user-centric. By selecting SLIs that capture performance aspects that are relevant to the user, you can better meet SLA needs.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Explain how IT operations need to adapt to thrive in the cloud

References:
[Services.google.com > Increasing Business Value with Better IT Operations: A guide to Site Reliability Engineering \(SRE\)](#)

Related Material

A company is consulting with you to quickly scale their existing system architecture, which utilizes virtual machines. You decide to recommend VMware Engine to them. Which of these will the company be able to achieve with this recommendation?

- A)** Perform replatforming of virtual machines without changing their location
- B) Perform monitoring, verification, and optimization tasks for a network**
- C)** Utilize a fully managed and highly available file service
- D)** Perform migration of existing virtual machines to new containers

Explanation

Using VMware Engine, the company can perform replatforming of virtual machines (VMs) without changing their location. With VMware Engine, you can move applications that are based on VMware onto Google Cloud without any changes to existing applications. It provides all the necessary infrastructure and VMware licenses that allow you to have a software-defined data center (SDDC) on Google Cloud. Using VMware Engine provides elasticity and scaling through the ability to swiftly build a private cloud using flexible management of resources and automatic scaling.

You cannot perform migration of existing VMs to new containers using VMware Engine. A key difference between containers and VMs is that containers perform virtualization at the level of the OS while VMs perform virtualization at the level of hardware. Containers are software packages that contain application code as well as dependencies like runtime environments and libraries that are required for running an application. This allows containers to virtualize an OS and allow applications to be made portable across multiple systems.

You cannot perform monitoring, verification, and optimization tasks for a network using VMware Engine. You can use Network Intelligence Center for that. Network Intelligence Center is used for monitoring, verifying, and optimizing a network and can be used for Google Cloud, on-premises, and multi-cloud systems. It provides a reduction in the risks of outages occurring and also ensures greater levels of compliance with regulations and security.

You cannot utilize a fully managed and highly available file service using VMware Engine. For that, you can use Cloud Filestore. You would use Cloud Filestore for enterprise-grade data applications that need storage with durability and persistence, which can be accessed using a Network File System (NFS). You can also use Cloud Filestore for storage systems that need a file system that is Portable Operating System Interface (POSIX) compliant.

Objective:
Infrastructure & Application Modernization

Sub-Objective:
Learn what modernizing IT infrastructure with Google Cloud means

References:
[Cloud.google.com > Google Cloud VMware Engine](#)

Related Material

You are a cloud services consultant and are asked to give guidance on adopting cloud technology to several interested companies. Which of these would you specify as examples of applications of machine learning (ML) services? (Choose two.)

- A)** Automation of the processing required for mortgage applications
- B)** Reporting of customer credit card chargebacks in a fraud report
- C) Provision of customer support through a dedicated customer support officer**
- D) Analysis of customer sentiments and sales forecasts**
- E)** Deployment of sales professionals for finding new customers

Explanation

Automation of the processing required for mortgage applications and analysis of customer sentiments and sales forecasts are two applications of machine learning (ML) services. Robotic process automation (RPA), when used in conjunction with ML, allows for creating automation systems that have the intelligence to perform tasks, such as processing mortgage applications. ML services can also be used to optimize sales by analyzing the sentiments of customers and performing predictions for customer churn. Additionally, ML systems can be used to increase the cybersecurity of an organization through intelligent threat monitoring to protect against hackers and malware.

Provision of customer support through a dedicated customer support officer is not an application of ML. This requires a person to perform various customer support tasks. When using ML for customer service tasks, you use virtual assistants and chatbots to automate routine customer service tasks. This can help accelerate the process of resolving customer issues as well.

Reporting of customer credit card chargebacks in a fraud report is not an ML process. Reporting of credit card data is simply reporting data. There is no ML application taking place for this. You would use an automated ML process to search through massive amounts of transaction data to identify any fraudulent activity in real-time.

Deployment of sales professionals for finding new customers is not an application of ML. This is a manual process. When using ML for marketing in the digital domain, you would use ML services to automate the process of searching for new customers and offering them the appropriate services in a timely manner.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Identify Google Cloud's solutions for machine learning and AI

References:
[Cloud.google.com > What is machine learning?](#)

Related Material

Your company uses Google's GKE for automatically managing and deploying container-based applications. As per the shared responsibility model, which of these is the responsibility of the customer?

- A)** Encryption of data at rest
- B) The operating system of nodes**
- C)** Workloads containing docker files
- D)** The control pane

Explanation

The customer is responsible for workloads that contain docker files, application code, container images, Identity and Access Management (IAM) and Role-Based Access Control (RBAC) policies, and pods. The customer is also responsible for the nodes that are running workloads. This also includes the software that may be installed on nodes as well as any changes to the configuration of the nodes. Node updating is also the responsibility of the customer.

The operating system of nodes, encryption of data at rest, and the control pane are the responsibility of Google. This also includes the cloud infrastructure, which comprises OS, networking, storage, firmware, and hardware. Google is also responsible for the encryption of data at rest and in transit. The operating system in use on nodes, which can be Ubuntu or Container-Optimized OS (COS). is the responsibility of Google. Google Kubernetes Engine (GKE) can automatically deploy upgrades to the images. This is separate from the OS that is running inside of containers. Also, GKE manages the control pane that comprises master virtual machines (VMs), IAM authentication systems, cluster certificate authority (CA), and the application programming interface (API) server. The necessary patching and upgrades needed for these components are supported by a service level objective (SLO). The control pane is the part of the Kubernetes system that controls how communication with a cluster is performed and how user states are implemented. Google hardens this control pane to protect against vulnerabilities.

Objective:
Understanding Google Cloud Security and Operations

Sub-Objective:
Describe a cloud security approach and Google Cloud security benefits

References:
[Cloud.google.com > Exploring container security: the shared responsibility model in GKE](#)

Related Material

Question #60 of 60

Question ID: 1495181

You work for a company that is exploring how Google Cloud can allow it to apply its data in new ways. The company wants its applications to be responsive to changing user needs, which can be serviced on demand. What aspect of Google Cloud would be useful in this scenario?

- A) Storage of data as archives for access over a lengthy period

✖
- B) Ingestion and analysis of user data performed in real-time

✔
- C) Removal of overhead related to managing data access

✖
- D) Validation and cleaning of data done automatically

✖

Explanation

Ingestion and analysis of user data performed in real-time is a distinct advantage of using Google Cloud that is relevant to this scenario. To be responsive to changing user needs and preferences, a real-time system is required to collect user data. Gathering and analyzing user data in real-time is done easily on a cloud deployment. Such real-time systems allow applications to make necessary immediate changes to address various user requirements dynamically based on ingested real-time information.

Storage of data as archives for access over a lengthy period is not a relevant advantage in this scenario. Archiving data is a benefit of using the cloud, but it is for data that does not need to be accessed immediately and is not suitable for real-time processing needs. Archiving is a way to keep low-cost storage for infrequently accessed data and meet regulatory and compliance requirements. Archiving using the cloud is not a way to implement real-time user interaction and data requirements.

Validation and cleaning of data done automatically is not a relevant advantage in this scenario. Automatic validation and cleaning of data are useful for applications that need quick and efficient data processing, such as machine learning (ML) applications that better serve users in an efficient and automated manner. The company takes responsibility for the quality and validity of the stored data. This does not represent the kind of real-time ingestion and analysis that is needed in this scenario.

Removal of overhead related to managing data access is not a relevant advantage in this scenario, which requires real-time data processing. Having a centralized Identity and Access Management (IAM) system using user access policies is a key advantage of using a cloud system. When an organization moves to a cloud deployment, it needs to take full responsibility for the access management of its data. That is not the responsibility of the cloud service provider (CSP), but the customer. The CSP looks after the security of the infrastructure that supports the cloud but not of data stored in the cloud by the customer.

Objective:
Innovating with Data and Google Cloud

Sub-Objective:
Describe the role of data in digital transformation and the importance of a data-driven culture

References:
[Cloud.google.com > Google's guide to building a data-driven culture](#)

Related Material